

# AMA 564 Deep Learning

2026 Spring

## Lecture 1

## Course Information

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| Start Week | End Week | Day of Week | Start Time | End Time | Venue  |
|------------|----------|-------------|------------|----------|--------|
| 1          | 13       | Fri         | 12:30      | 15:20    | SHA030 |

## Deep Neural Networks

- Multi-Layer Perceptron
- Convolutional Neural networks
- Recurrent Neural Networks
- Transformer

## Learning Algorithms

- Regression, classification
- Classification
- Generative Learning
- Vision
- Natural Language Processing

Backpropagation

Optimization

Overview of Training LLMs



| Week | Date   | Content                                                    |
|------|--------|------------------------------------------------------------|
| 1    | Jan 16 | Multilayer Perceptrons I: History and Development          |
| 2    | Jan 23 | Multilayer Perceptrons II: Regression and Training         |
| 3    | Jan 30 | Backpropagation and Optimization                           |
| 4    | Feb 06 | Computer Vision I: Convolution Neural Networks             |
| 5    | Feb 13 | Computer Vision II: Architectures and Training             |
| 6    | Feb 27 | Generative Models I: VAE                                   |
| 7    | Mar 06 | Generative Models II: GAN and others                       |
| 8    | Mar 13 | Natural Language Processing I: Recurrent Neural Networks   |
| 9    | Mar 20 | Natural Language Processing II: Attention-based Models     |
| 10   | Mar 27 | Natural Language Processing III: Large Models and Training |
| 11   | Apr 03 | General holiday (Good Friday)                              |
| 12   | Apr 10 | Natural Language Processing IV: Fine-tuning and RLHF       |
| 13   | Apr 17 | Overall Course Review                                      |



## After this course, you will

- Know different deep neural networks  
(can tell MLP, CNN, RNN, etc.)
- Understand how to train deep neural networks  
(backpropagation, optimization techniques)
- Apply or design algorithms for different learning tasks  
(regression, classification, generation; vision, natural language processing)
- Implement data analysis or learning task on python in practice  
(deep learning with PyTorch for conventional learning tasks)
- Know the key steps in building LLMs.

## Prerequisite

- Linear Algebra  
Basic Calculus  
Probability and statistics
- Python Basics

## 13 Weeks

- **Lecture 2** hours each week  
**Tutorial/Demo 1** hour each week
- **Homework (30%)**       $\leq$       **3**  
**Project (20%)**          =          **1**
- **Final (50%)**

## References

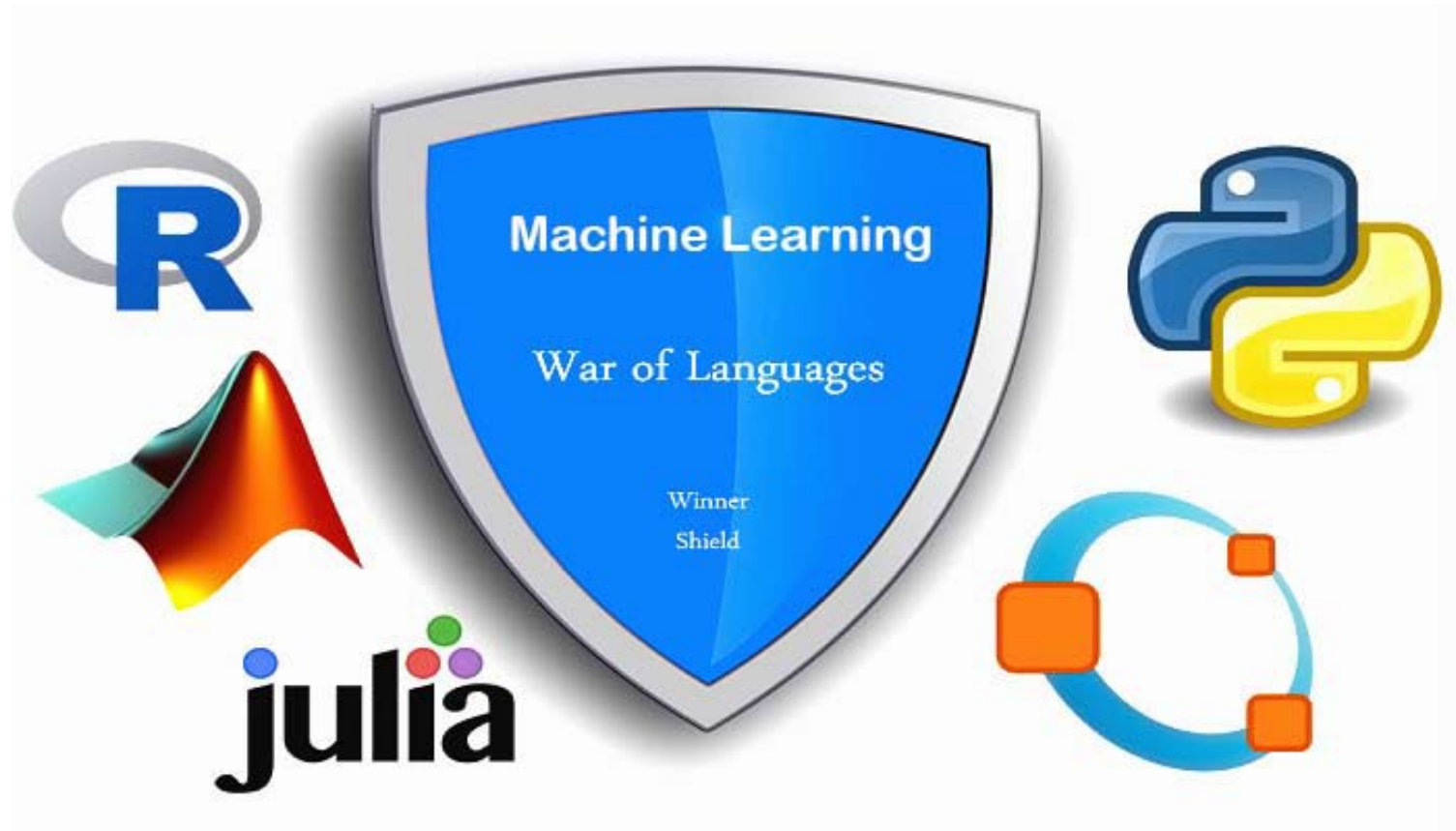
1. *Deep Learning*. Goodfellow I., Bengio Y., Courville A. The MIT Press 2016.
2. *Deep Learning with Python*. Francois Chollet with J. J. Allaire. Manning Publications Co.2018
3. *Foundations of Machine Learning*. Mohri, M., Rostamizadeh, A., & Talwalkar, A. The MIT Press 2018.
4. *The Elements of Statistical Learning*. Hastie, T., Tibshirani, R, and Friedman, J. Springer 2009.
5. *Convex optimization: Algorithms and complexity*. Bubeck, S. Now Publishers Inc.2015

## Rubrics (**Waiting for Approve from Department**)

| Specific assessment methods/tasks | % weighting |
|-----------------------------------|-------------|
| Assignments                       | 30%         |
| Projects                          | 20%         |
| Midterm Test                      | None        |
| Examination                       | 50%         |
| <b>Total</b>                      | <b>100%</b> |

| Grade Level | Cut-off       |
|-------------|---------------|
| A-/A/A+     | 85            |
| B-/B/B+     | 70            |
| C-/C/C+     | 55            |
| D/D+        | 40            |
| <b>F</b>    | <b>&lt;40</b> |

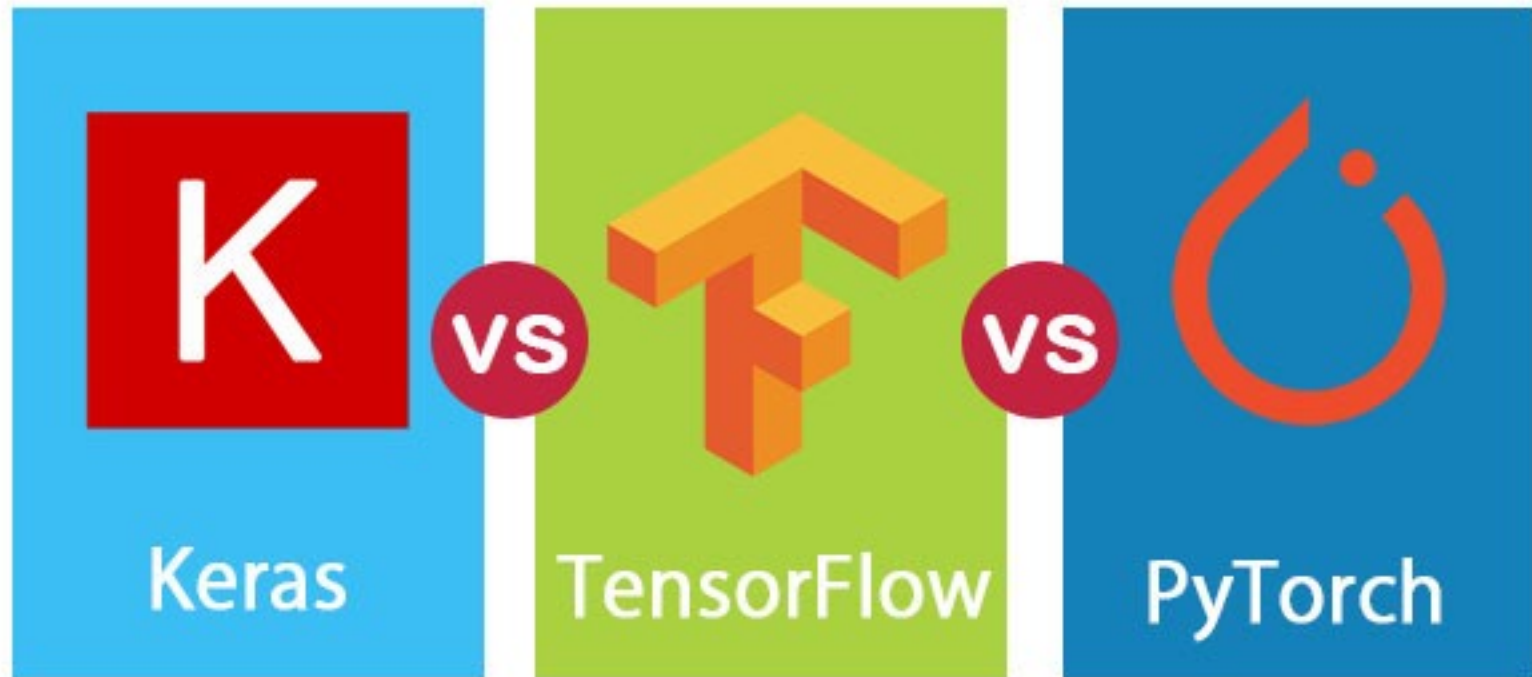
## Programming



Source: [https://www.codementor.io/@innat\\_2k14/preferable-tools-for-machine-learning-python-matlab-r-jfozzpphz](https://www.codementor.io/@innat_2k14/preferable-tools-for-machine-learning-python-matlab-r-jfozzpphz)

In this course, we use **Python**.

## Programming



Source: <https://www.educba.com/keras-vs-tensorflow-vs-pytorch/>

In this course, we use **PyTorch**.

## Programming



<https://pytorch.org/>

PyTorch

Get Started

Ecosystem ▾

Mobile

Blog

Tutorials

Docs ▾

Resources ▾

GitHub

Q

Shortcuts

Overview

PyTorch 2.x: faster, more pythonic and as

dynamic as ever

Testimonials

Motivation

Technology Overview

User Experience

Distributed

Developer/Vendor Experience

Final Thoughts

Accelerating Hugging Face And Timm

Models With Pytorch 2.0

Requirements

Getting Started

FAQs

Ask The Engineers 2.0 Live Q&A Series

series better and thank you all who have made the 1-series so successful.

### PYTORCH 2.X: FASTER, MORE PYTHONIC AND AS DYNAMIC AS EVER

Today, we announce `torch.compile`, a feature that pushes PyTorch performance to new heights and starts the move for parts of PyTorch from C++ back into Python. We believe that this is a substantial new direction for PyTorch – hence we call it 2.0. `torch.compile` is a fully additive (and optional) feature and hence 2.0 is 100% backward compatible by definition.

Underpinning `torch.compile` are new technologies – TorchDynamo, AOTAutograd, PrimTorch and TorchInductor.

- **TorchDynamo** captures PyTorch programs safely using Python Frame Evaluation Hooks and is a significant innovation that was a result of 5 years of our R&D into safe graph capture
- **AOTAutograd** overloads PyTorch's autograd engine as a tracing autodiff for generating ahead-of-time backward traces.
- **PrimTorch** canonicalizes ~2000+ PyTorch operators down to a closed set of ~250 primitive operators that developers can target to build a

## Programming

Tutorials: <https://pytorch.org/tutorials>

Tutorials > Welcome to PyTorch Tutorials



### Learn the Basics

A step-by-step guide to building a complete ML workflow with PyTorch.

Getting Started



### Introduction to PyTorch on YouTube

An introduction to building a complete ML workflow with PyTorch. Follows the PyTorch Beginner Series on YouTube.

Getting Started



### Learning PyTorch with Examples

This tutorial introduces the fundamental concepts of PyTorch through self-contained examples.

Getting Started



Examples





## Programming

## Helpful Documents:

<https://pytorch.org/docs/stable/index.html>



Get Started

Ecosystem

Mobile

Blog

Tutorials

Docs ▾

Resources ▾

GitHub

1.13 ▼

🔍 tensor

Community [ + ]

Developer Notes [ + ]

Language Bindings [ + ]

Python API [ - ]

torch

torch.nn

torch.nn.functional

torch.Tensor

Tensor Attributes

Tensor Views

torch.amp

torch.autograd

torch.library

torch.cuda

torch.backends

torch.distributed

torch.distributed.algorithms.join

torch.distributed.elastic

Docs > Search

### Searching...

- `torch.jit.unsupported_tensor_ops` (Python module, in TorchScript Unsupported Pytorch Constructs)
- `torch.masked.maskedtensor` (Python module, in torch.masked)
- `torch.utils.tensorboard` (Python module, in torch.utils.tensorboard)
- `torch.tensor` (Python function, in torch.tensor)
- `torch.Tensor` (Python class, in torch.Tensor)
- **Autograd mechanics**  
...rations that created  
the data as you execute operations, giving you a directed acyclic graph  
whose leaves are the input tensors and roots are the output tensors.  
by tracing this graph from roots to leaves, you can automatically  
compute the...
- **Automatic differentiation package - torch.autograd**  
...ation of arbitrary scalar valued functions. it requires minimal  
changes to the existing code - you only need to declare tensor s  
for which gradients should be computed with the `requires_grad=True` keyword.  
as of now, we only support autograd...
- **C++**  
...le model modifications if needed (e.g. pulling out submodules)

## Tutorial



## Personal or Course Project



# 2023: A Big Year for Artificial Intelligence



ChatGPT

chat.openai.com/chat

Reset Thread

Light Mode

OpenAI Discord

Learn More

Log out

This causes the component to re-render, and the updated `count` value is displayed in the `p` element.

I hope this helps! Let me know if you have any other questions.

Can you add App.tsx that uses this component?

Sure! Here's an example of an `App` component that uses the `Counter` component from the previous example:

Copy code

```
import React from 'react';
import Counter from './Counter';

function App() {
  return (
    <div>
      <Counter />
    </div>
  );
}
```

In this example, the `App` component simply renders the `Counter` component. When the `Counter` component is rendered, it will display a button and a paragraph that shows the current count. When the button is clicked, the `Counter` component will update its state and re-render, showing an updated count in the paragraph.

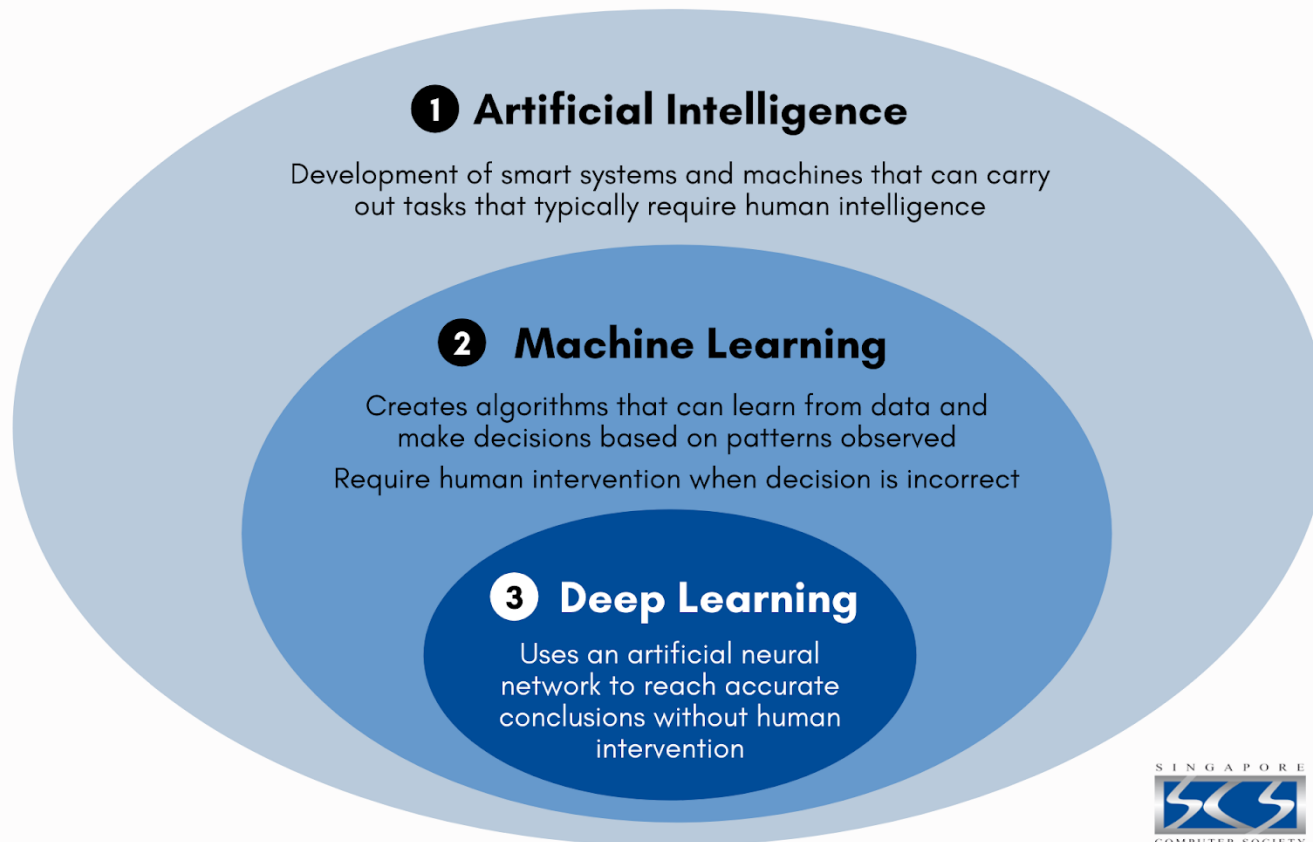
Free Research Preview: ChatGPT is optimized for dialogue. Our goal is to make AI systems more natural to interact with, and your feedback will help us improve our systems and make them safer.

# A branch of powerful LLMs



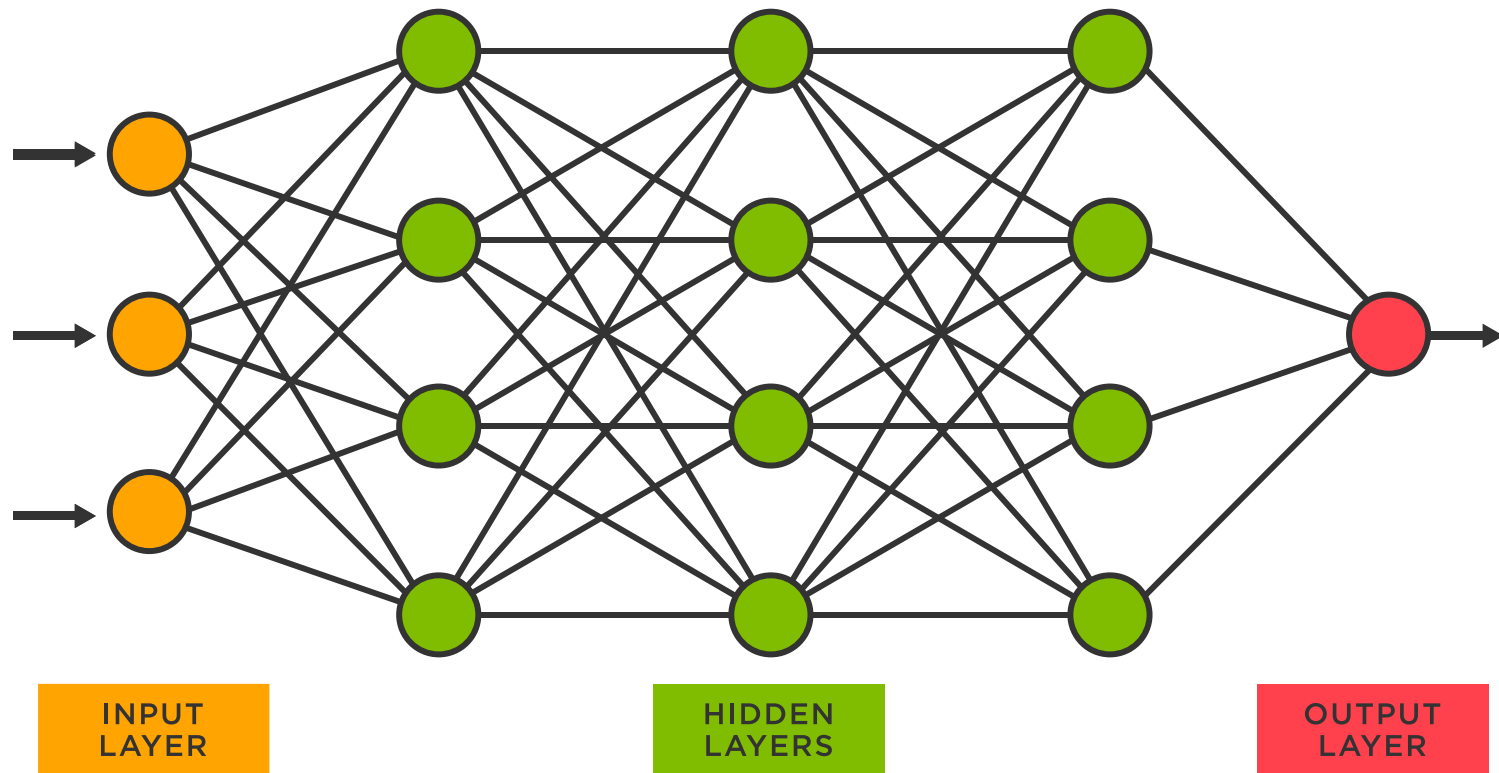
|                                                                                                    |                                                                                                   |                                                                                                       |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  <b>ChatGPT</b>    |  <b>Claude</b>   |  <b>Gemini</b>     |
|  <b>Meta</b>       |  <b>DeepSeek</b> |  <b>Perplexity</b> |
|  <b>Midjourney</b> |  <b>Flux</b>     |  <b>Ideogram</b>   |
|  <b>Runway</b>    |  <b>Luma</b>    |  <b>Grok</b>      |

## ARTIFICIAL INTELLIGENCE VS MACHINE LEARNING VS DEEP LEARNING



**Deep learning** is part of a broader family of machine learning methods based on artificial neural networks with representation learning.

# What is Deep Neural Networks?



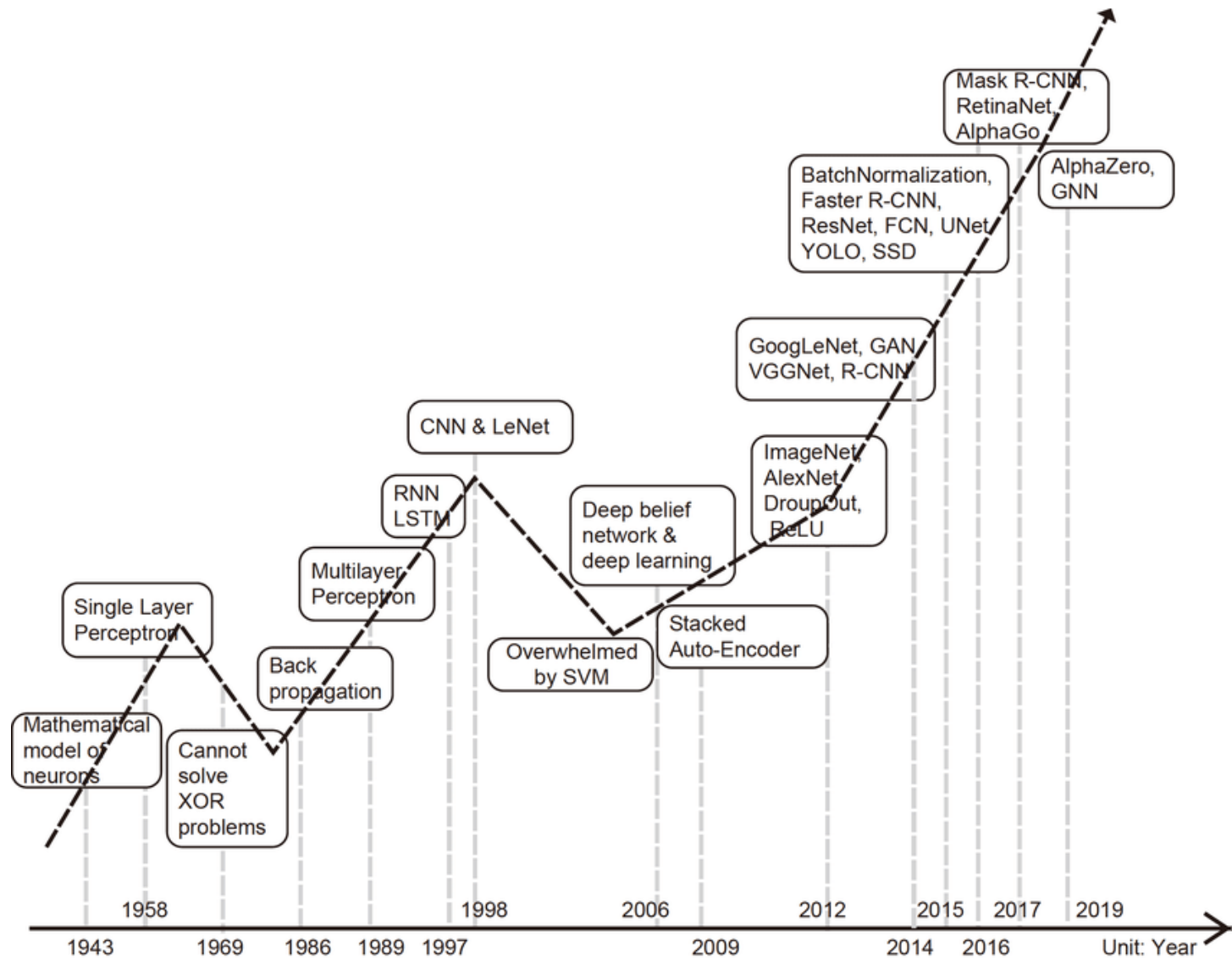
Source: <https://www.tibco.com/reference-center/what-is-a-neural-network>

Deep Learning utilizes **Artificial Neural Networks**

# A brief History of Artificial Intelligence



## A Three Wave Development of Deep Learning (Deep learning. Goodfellow, Bengio & Courville.)

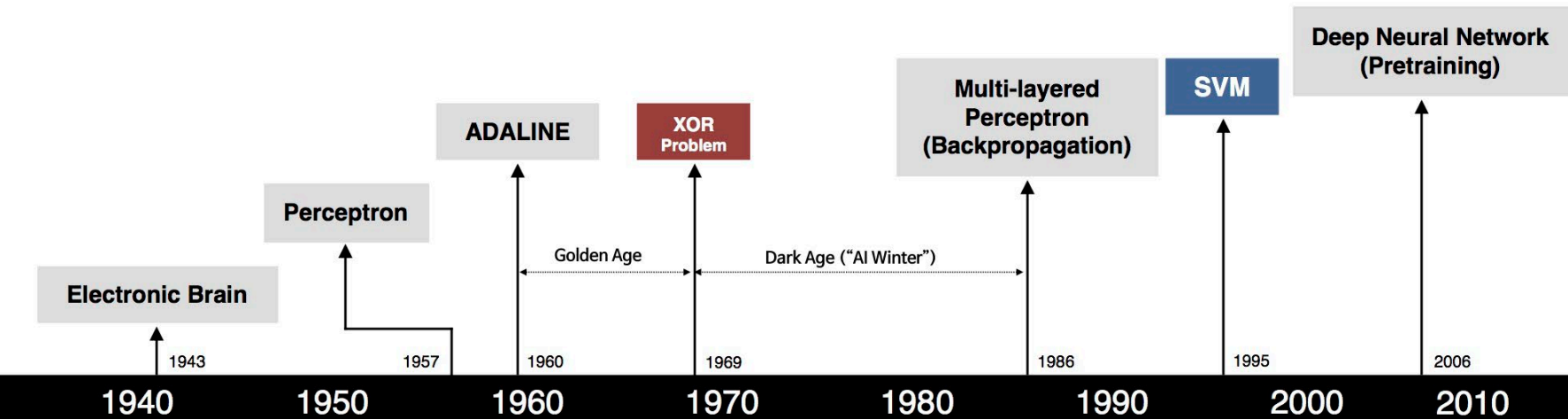


Source: "Application of deep learning in ecological resource research: Theories, methods, and challenges"

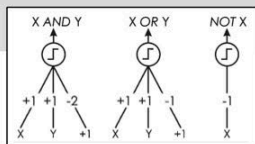
# A brief History of Artificial Intelligence



## Milestones



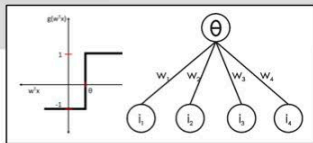
S. McCulloch - W. Pitts



- Adjustable Weights
- Weights are not Learned



F. Rosenblatt



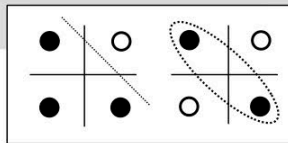
- Learnable Weights and Threshold



B. Widrow - M. Hoff



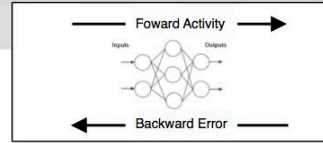
M. Minsky - S. Papert



- XOR Problem



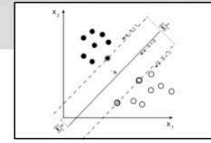
D. Rumelhart - G. Hinton - R. Williams



- Solution to nonlinearly separable problems
- Big computation, local optima and overfitting



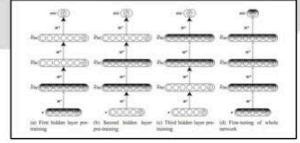
V. Vapnik - C. Cortes



- Limitations of learning prior knowledge
- Kernel function: Human Intervention



G. Hinton - S. Ruslan



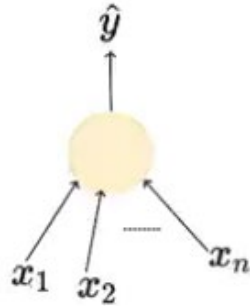
- Hierarchical feature Learning

Source: [http://beamlab.org/deeplearning/2017/02/23/deep\\_learning\\_101\\_part1.html](http://beamlab.org/deeplearning/2017/02/23/deep_learning_101_part1.html)



## The first Neuron 1943

MP Neuron



$$\hat{y} = 1 \text{ if } \sum_{i=1}^n x_i \geq b$$

$$\hat{y} = 0 \text{ otherwise}$$



Boolean inputs



Linear

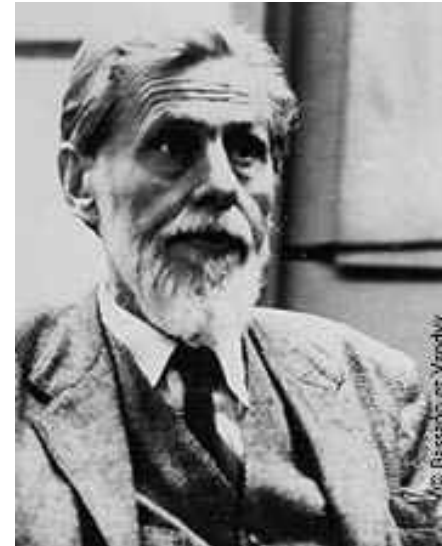


Inputs are not weighted



Adjustable threshold

**McCulloch-Pitts Neuron**



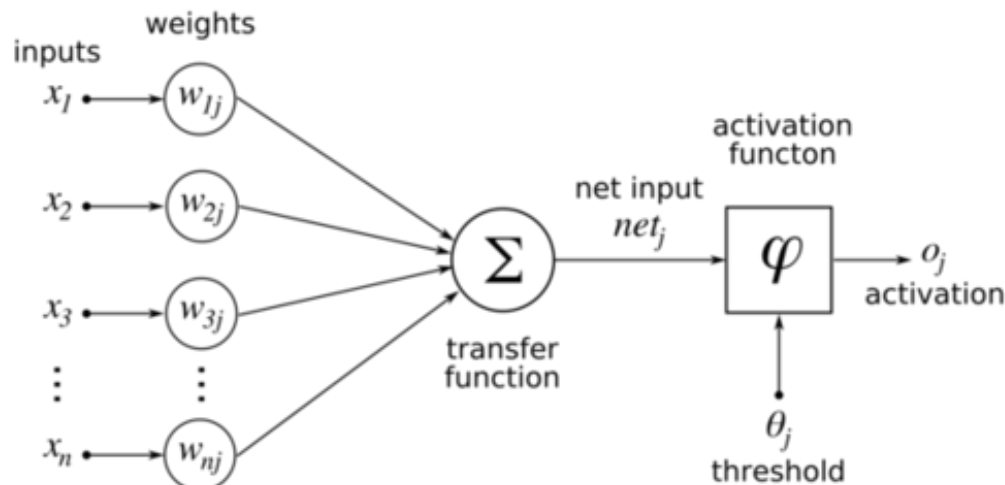
*Warren McCulloch*  
(Neuroscientist)



*Walter Pitts* (Logician)

## The first wave: 1940s-1960s

One of the very first ideas in machine learning and artificial intelligence



Rosenblatt's perceptron

- Date back to 1940s
- Could recognize letters and numbers
- Repeated promises of "True AI" that were unfulfilled

*"[The perceptron is] the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence."*

- Frank Rosenblatt, 1958



### Frank Rosenblatt

(July 11, 1928 – July 11, 1971)

American psychologist

"Father of deep learning"

(Tappert, Charles C. (2019).

["Who Is the Father of Deep Learning?"](#))

# A brief History of Deep Learning



The first wave: 1940s-1960s



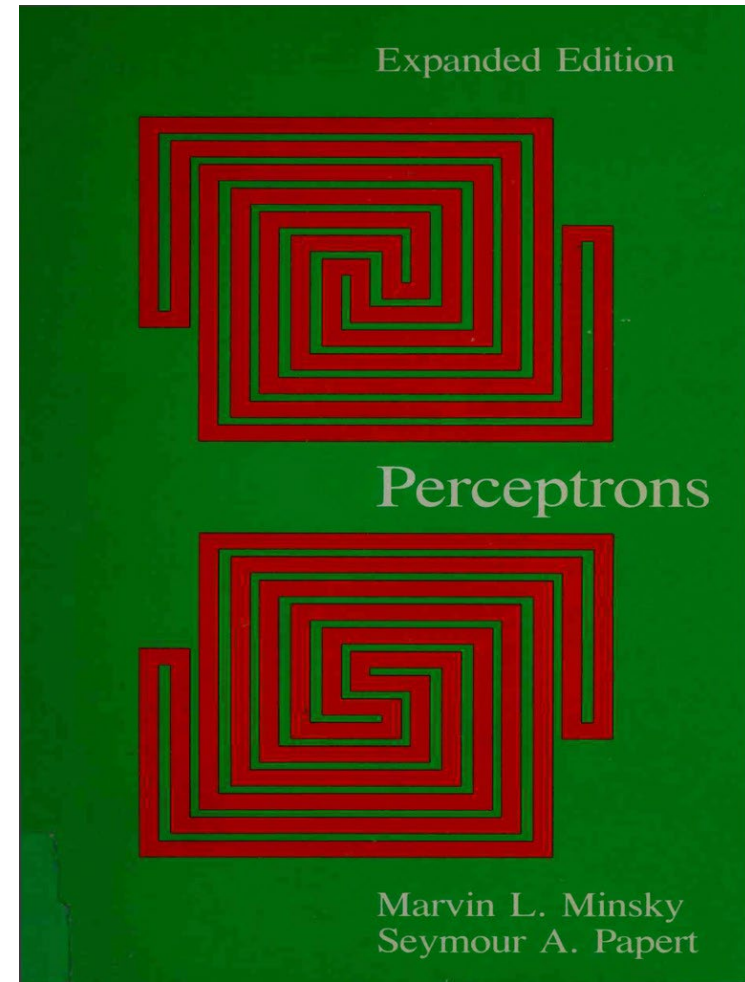
## The first AI Winter 1969



**Marvin Minsky** (left) and **Seymour Papert** (right) in 1971.  
*Cynthia Solomon/Courtesy of MIT*

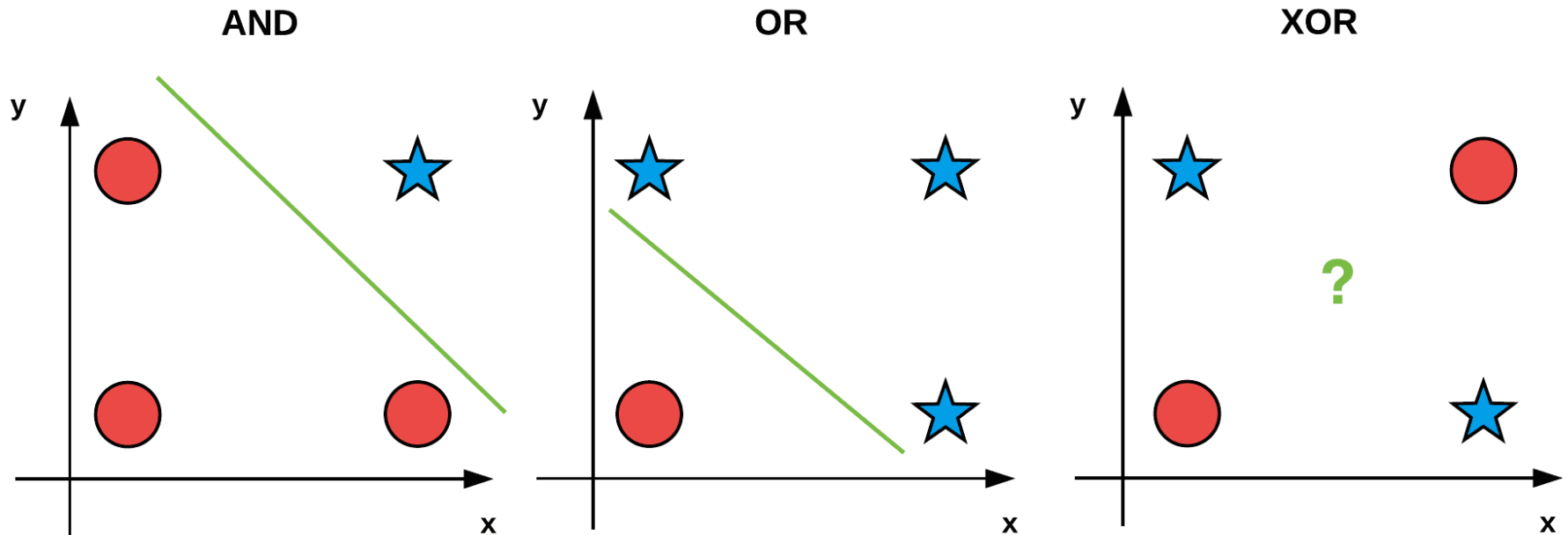
Source: <https://www.npr.org/sections/ed/2016/08/05/488669276/remembering-a-thinker-who-thought-about-thinking>

- They proved that perceptron can't even solve the XOR problem.
- End Rosenblatt's research on perceptrons when he was only 41 years old.
- Kills research on neural nets for the next 15-20 years.



The book "**Perceptron**"

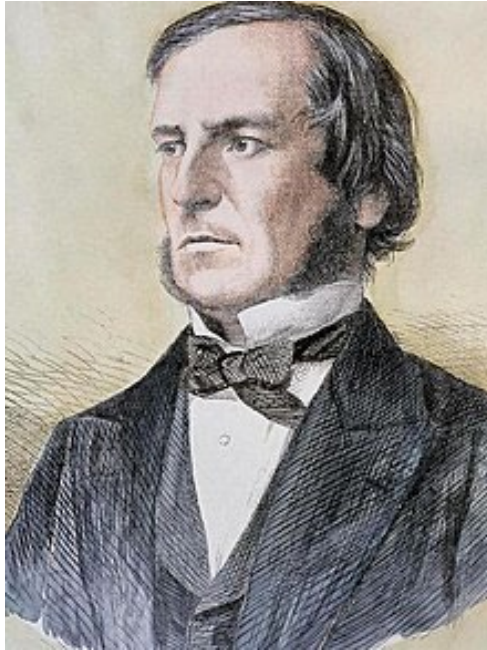
## The first AI Winter 1969



Source: <https://etaxfree.xyz/index.php?route=product/category&cid=28&cname=xor+deep+learning>

Linear Classifier can not do XOR.

## The Backpropagandists Emerge 1986



**George Boole**

2 November 1815 – 8 December 1864

Mathematician

Logician



**Geoffrey Hinton**

born 6 December 1947

Cognitive psychologist  
and computer scientist

**Great-great-grandson**



## The Backpropagandists Emerge 1986

- Along with David Rumelhart and Ronald Williams, Hinton published a paper entitled “Learning representations by back-propagating errors”.
- Neural nets with many hidden layers could be effectively trained by a relatively simple procedure.
- Additional layers endowed the network with the ability to learn nonlinear functions.

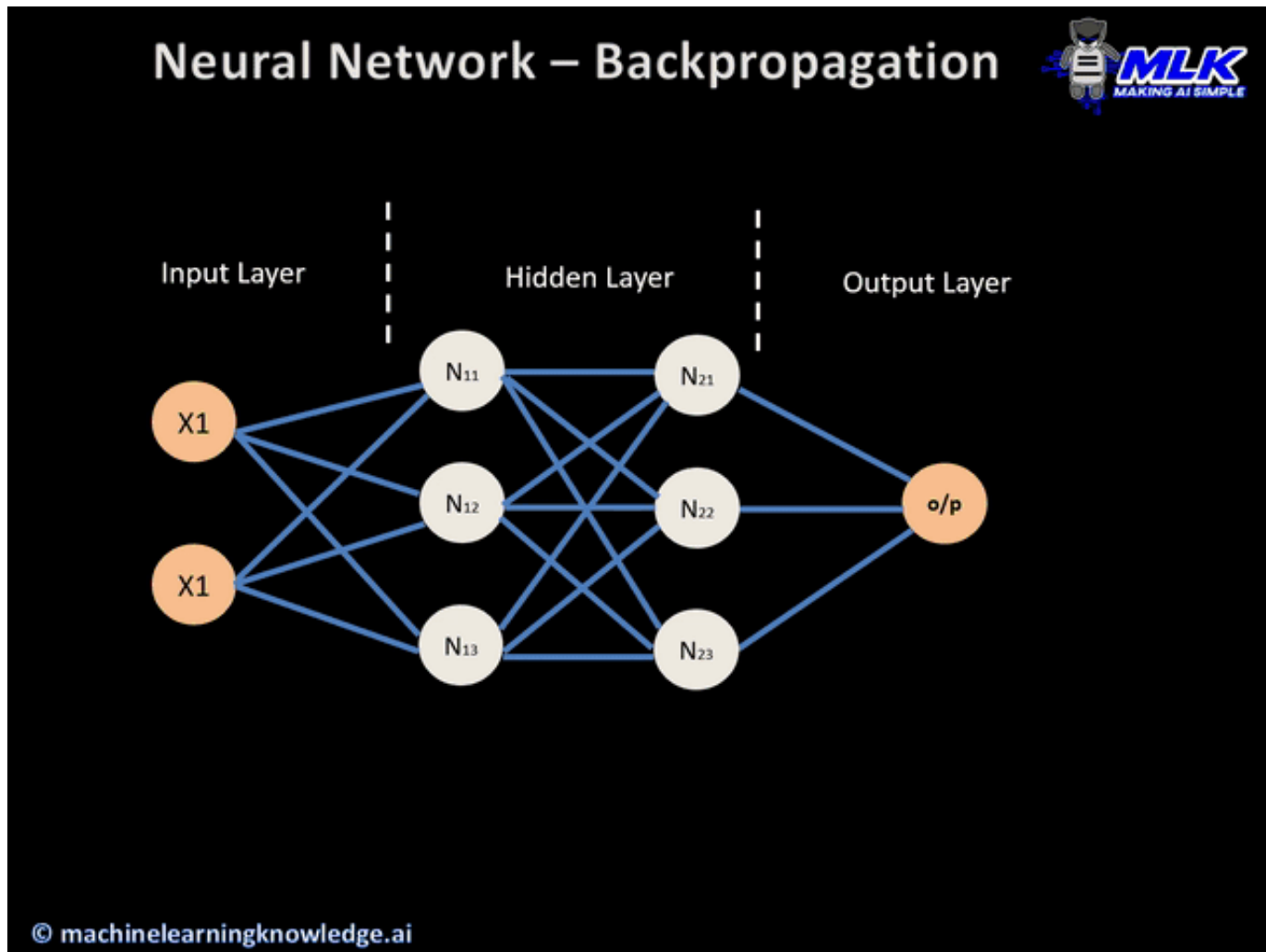


**Geoffrey Hinton**

born 6 December 1947  
Cognitive psychologist  
and computer scientist

Disputes: Firstly published in 1970 by Finnish master student **Seppo Linnainmaa**.

## The Backpropagandists Emerge 1986



Source: <https://medium.com/analytics-vidhya/backpropagation-for-dummies-e069410fa585>



## The Convolutional network & LeNet 1998



**Yann LeCun**

born 8 July 1960  
French computer scientist  
A founding father of  
convolutional nets

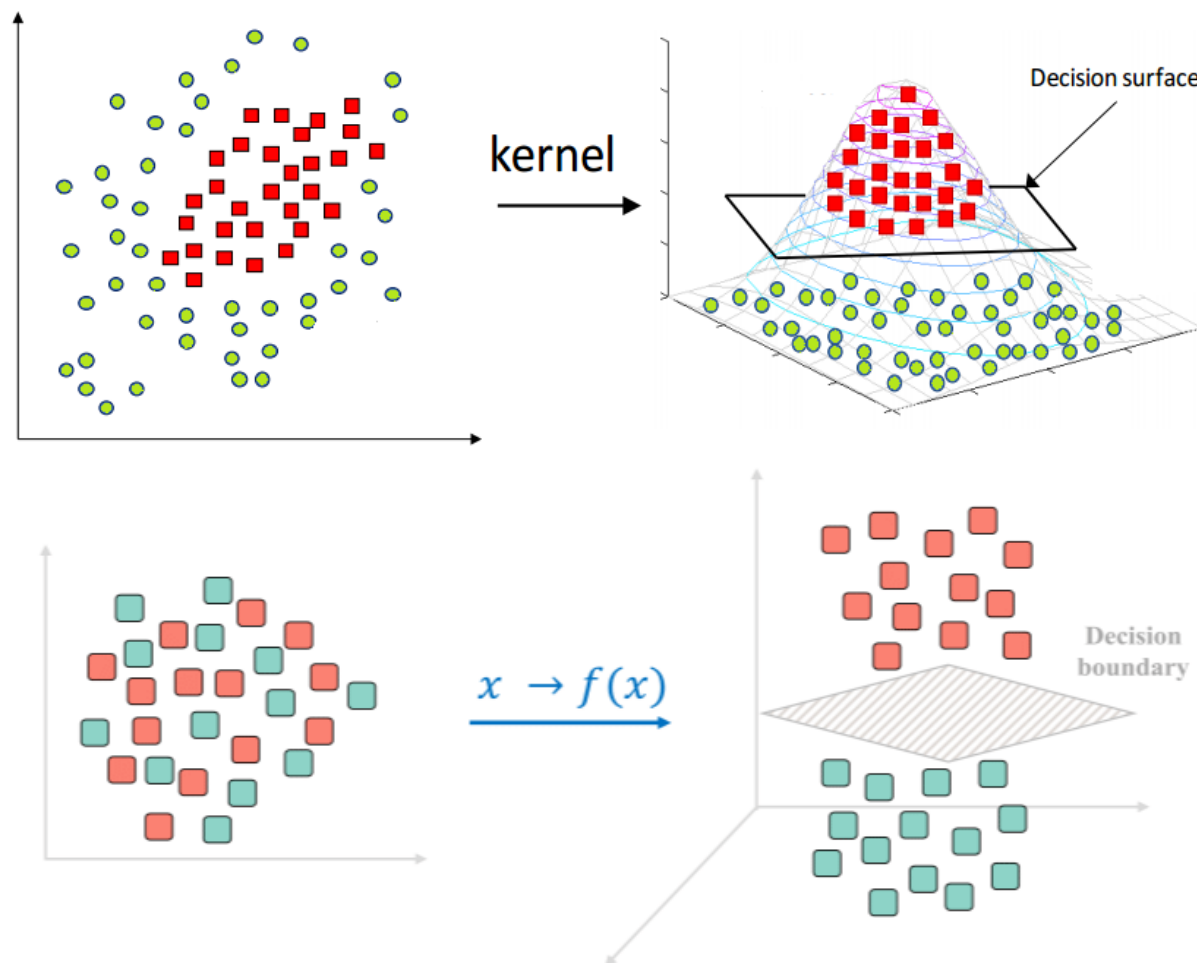
A project which was spearheaded by Yann Lecun at AT&T Bell Labs.

[LeNet-5 on MNIST data.](http://yann.lecun.com/exdb/lenet/dancing384.html)

Source: <http://yann.lecun.com/exdb/lenet/dancing384.html>

## The second AI Winter 1990s

- The Network approach didn't scale to larger problems.
- By the 90s the support vector machine (SVM) became the method of choice, and neural nets were moved back into storage.

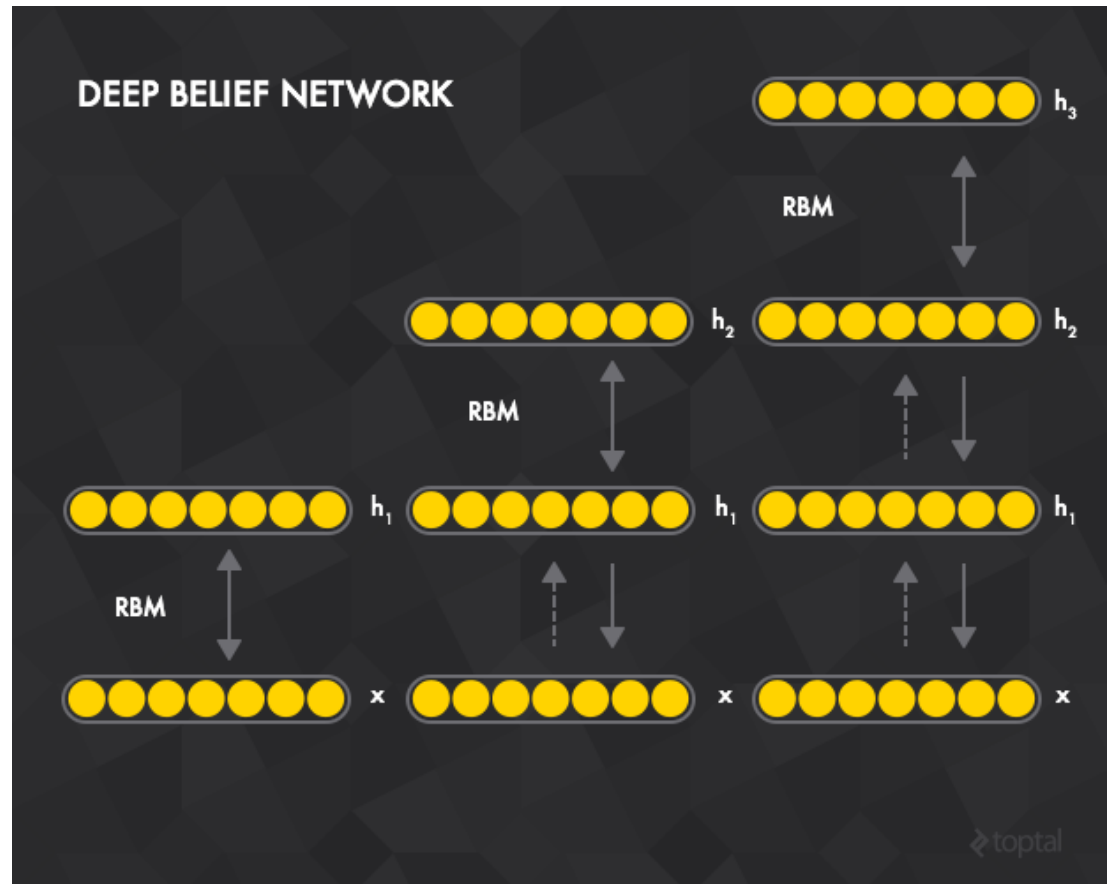


## Support Vector Machines

Source: Support Vector Machine vs. Random Forest for Remote Sensing Image Classification: A Meta-analysis and systematic review. Mohammadreza Sheykhoumoua and Masoud Mahdianpari.

## Rebranding as 'Deep Learning' 2006

- Hinton once again declared that he knew how the brain works.
- Unsupervised pre-training of "deep belief nets" allowed for large and deeper models.



Source: <https://www.toptal.com/machine-learning/an-introduction-to-deep-learning-from-perceptrons-to-deep-networks>

## The breakthrough 2012

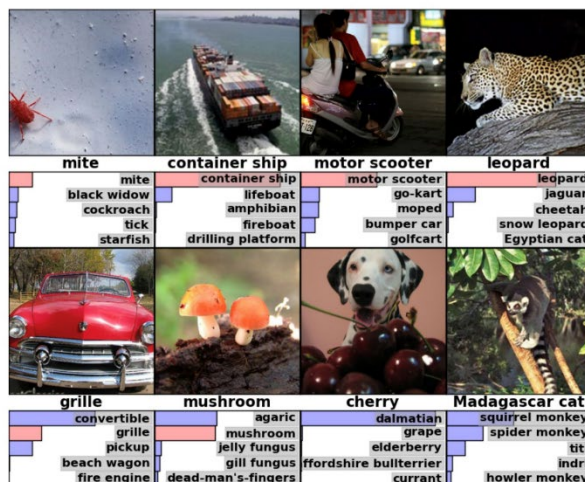
In 2010, a large database known as Imagenet containing millions of labelled images was created and published by Fei-Fei Li's group at Stanford.

This database was coupled with the annual LSVRC, where contestants would build computer vision models, submit their predictions, and receive a score based on how accurate they were.

## ImageNet Challenge

IMAGENET

- 1,000 object classes (categories).
- Images:
  - 1.2 M train
  - 100k test.



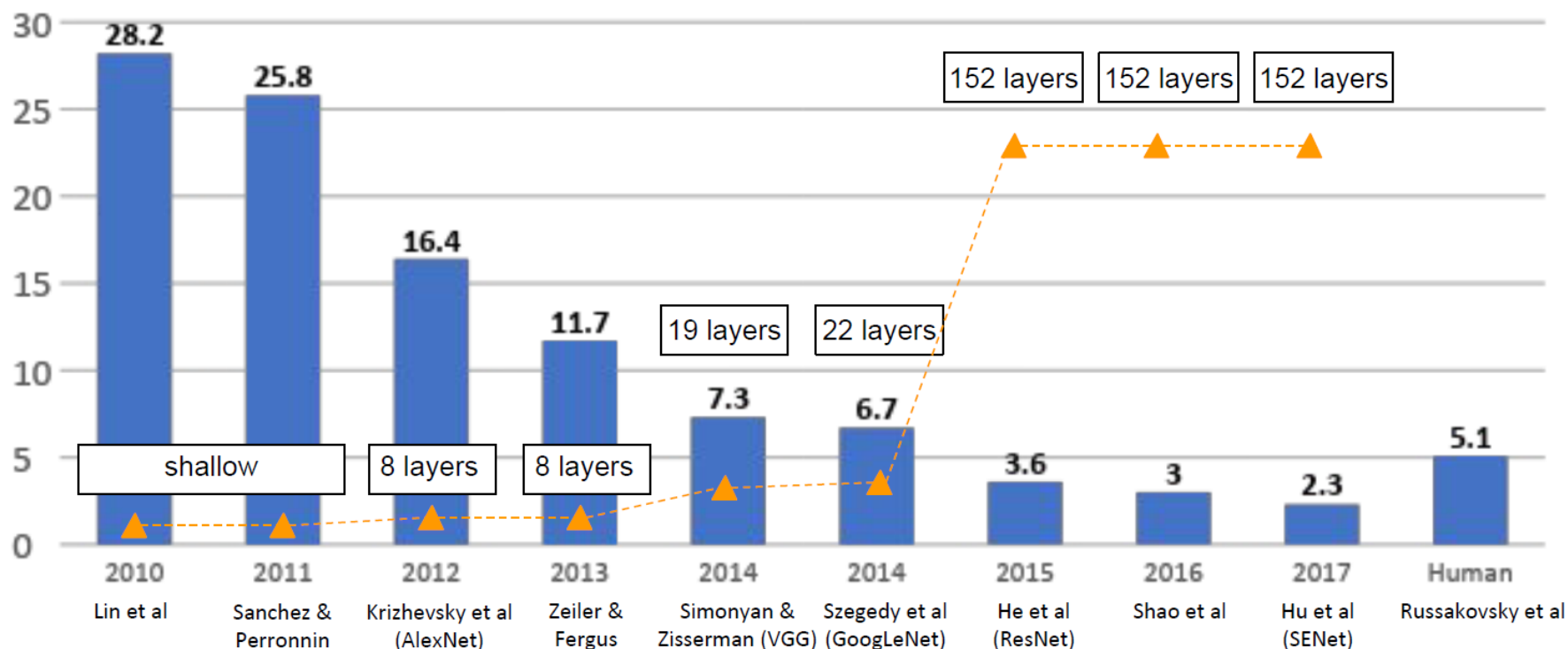
**Fei-Fei Li**

Computer Science Department at Stanford University  
Co-Director of Stanford's Human-Centered AI Institute

Homepage: <https://profiles.stanford.edu/fei-fei-li/>

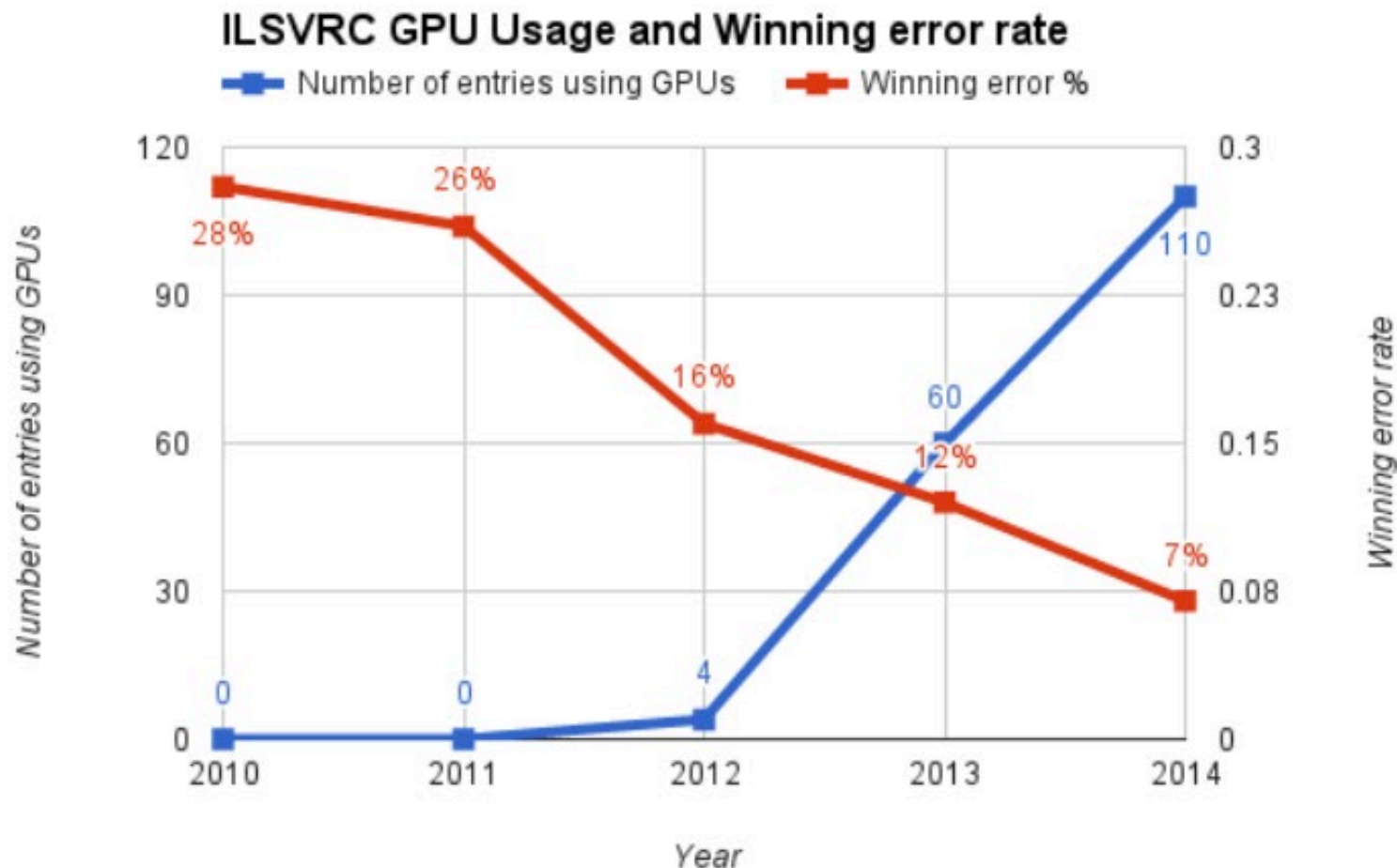
## Winners of ILSVRC

### ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



Source: [https://kjhov195.github.io/2020-02-10-CNN\\_architecture\\_2/](https://kjhov195.github.io/2020-02-10-CNN_architecture_2/)

## The hardware GPUs can help



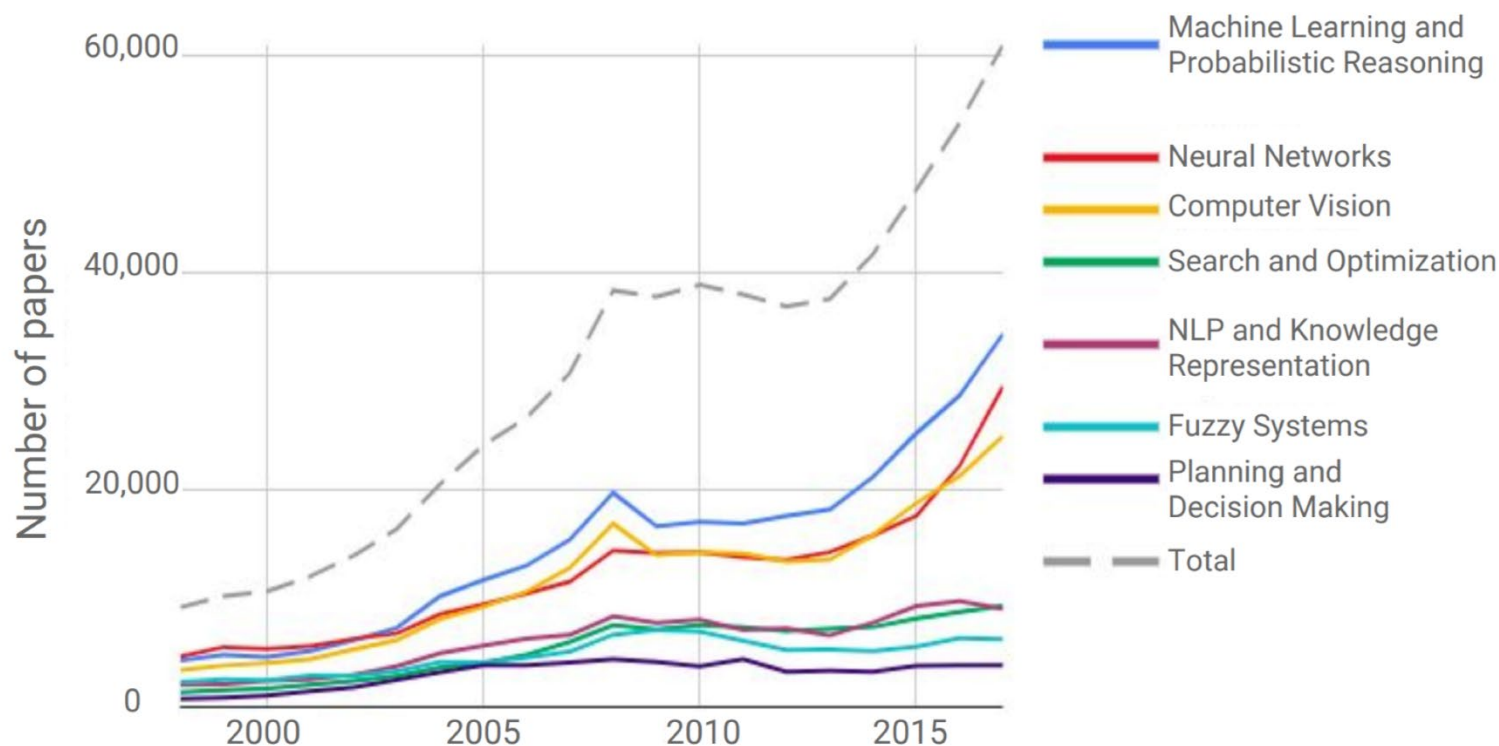
Here is a graph showing the error rate on Imagenet over time, note the precipitous drop in 2012 and the huge uptick in teams using GPUs.



## The breakthrough 2012

Number of AI papers on Scopus by subcategory (1998–2017)

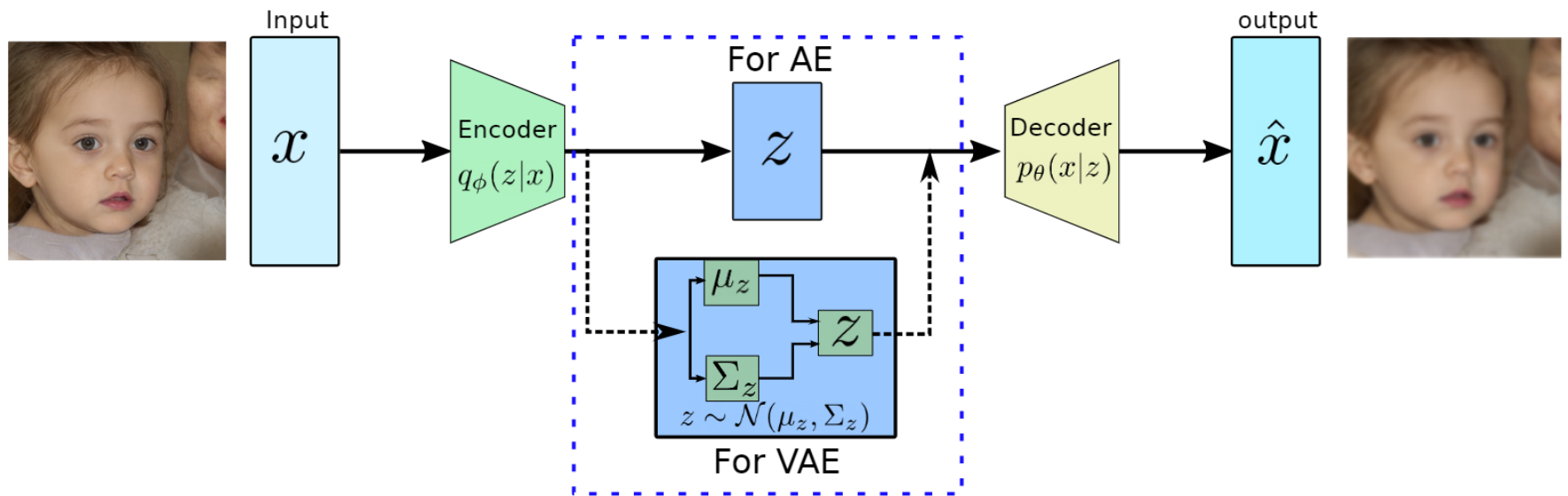
Source: Elsevier



*Neural networks are the second best way to do almost anything.*

--J.S. Denker

## Development of Deep Learning 2013



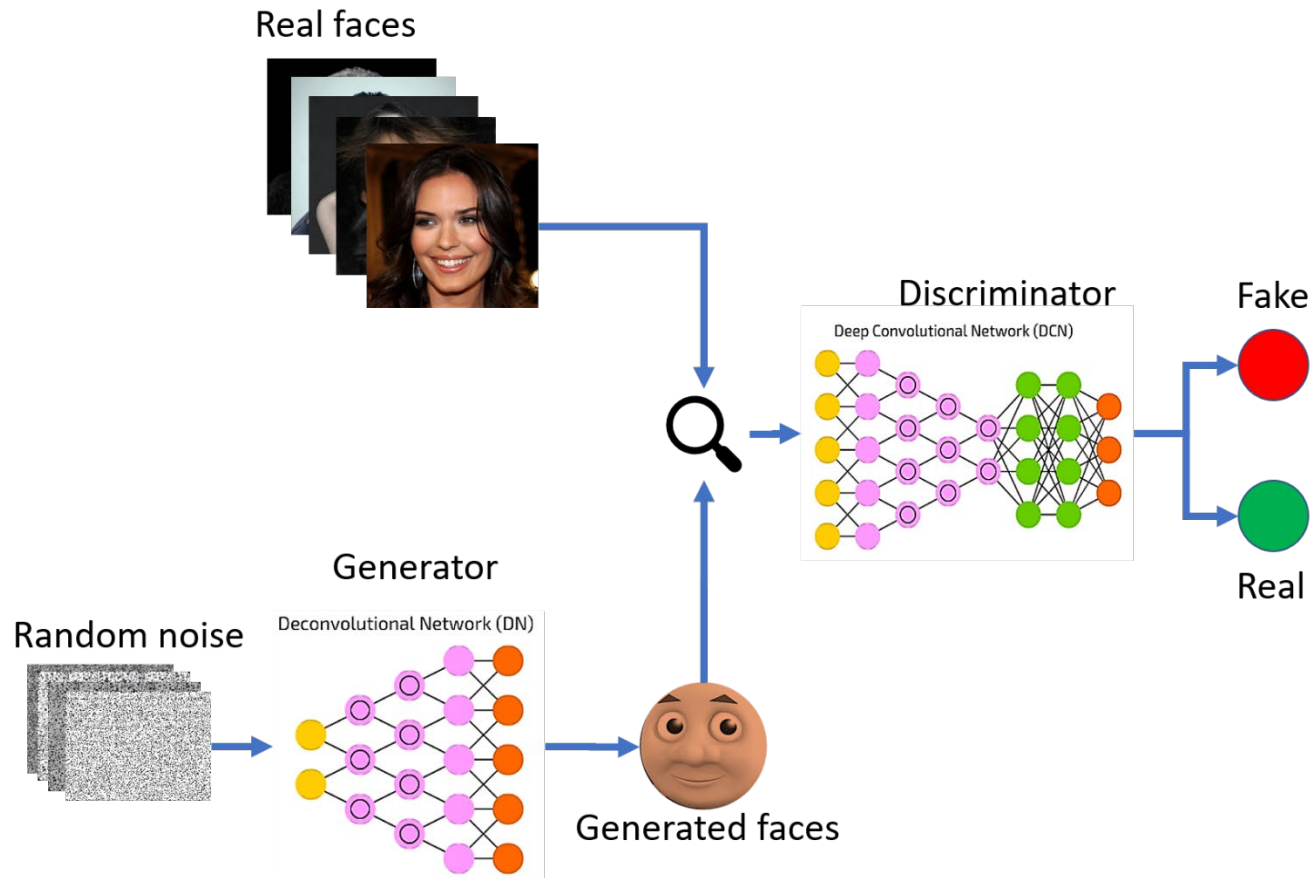
### Variational Auto-encoder

Auto-Encoding Variational Bayes (2013). Diederik P Kingma, Max Welling. <https://arxiv.org/abs/1312.6114>.

Cited by 24417



## Developing of Deep Learning 2014



### Generative Adversarial Networks

**Generative Adversarial Networks (2014).** Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio. <https://arxiv.org/abs/1406.2661>.

Cited by 53250

## 2018 Turing Award



**Fathers of the Deep Learning Revolution Receive ACM A.M. Turing Award.**

Bengio, Hinton and LeCun Ushered in Major Breakthroughs in Artificial Intelligence

# A brief History of Deep Learning



## 2022 OpenAI ChatGPT



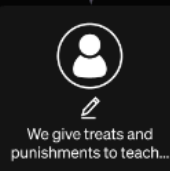
### Step 1

Collect demonstration data and train a supervised policy.

A prompt is sample from our prompt dataset.



A labeler demonstrates the desired output behavior.



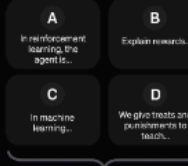
This data is used to fine-tune GPT-3.5 with supervised learning.



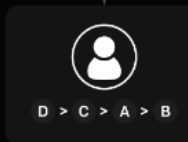
### Step 2

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



### Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

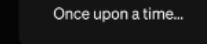
A new prompt is sampled from the dataset.



The PPO model is initialized from the supervised policy.



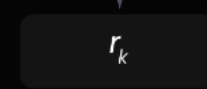
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



## 2024 Nobel Prize





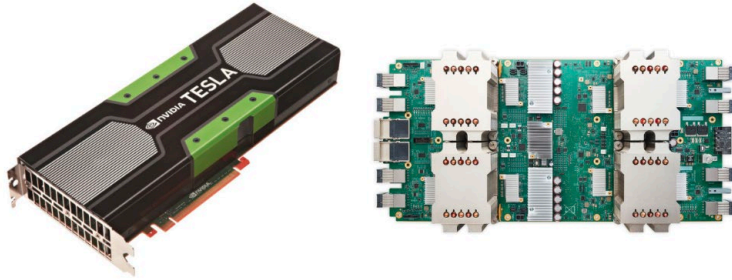
## 2025 Turing Award



# Why Deep Learning Now?



## Computing Power (GPU, TPU,...)



GPU and TPU

## Big Data



## Open Source Tools



PYTORCH



Microsoft  
CNTK

Caffe2

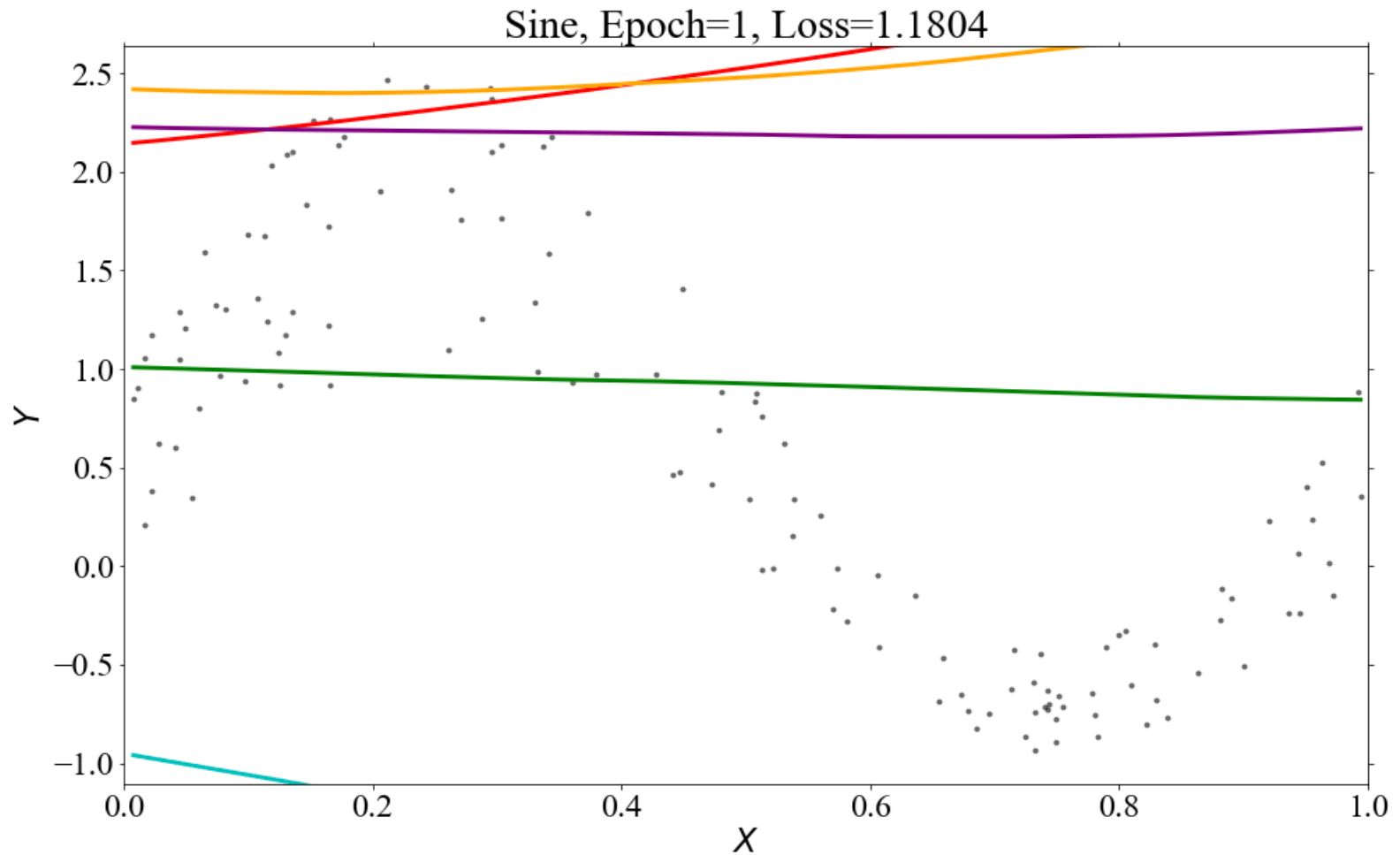
dmlc  
mxnet

gensim

spaCy

theano

## Regression

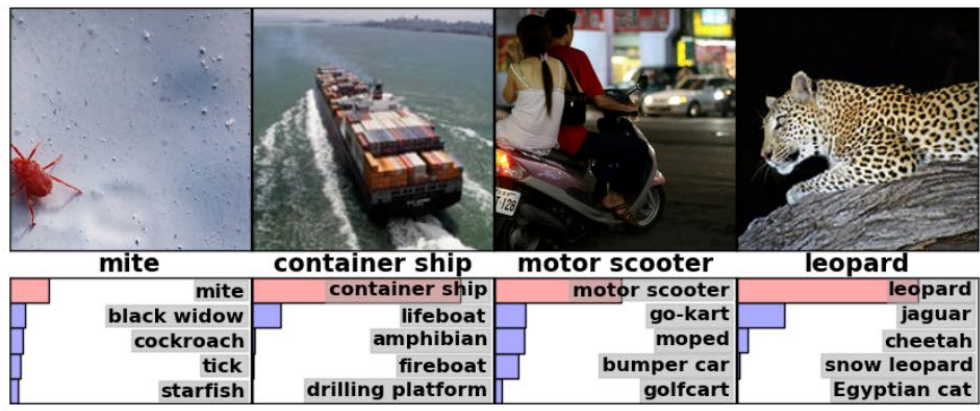


Least squares regression, quantile regression, robust regression...



# What can Deep Learning do?

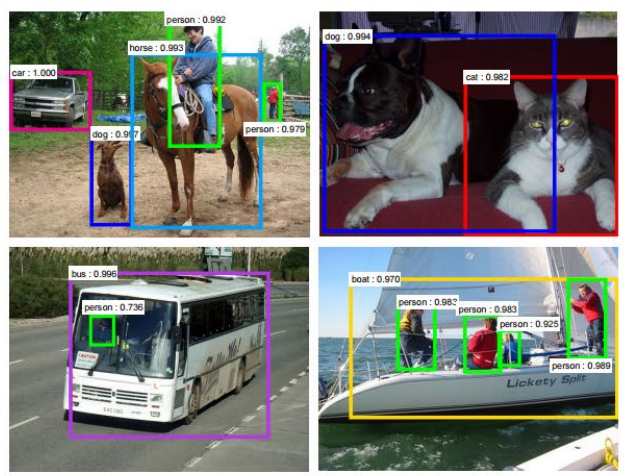
## Image recognition



[Krizhevsky 2012]



[Ciresan et al. 2013]



[Faster R-CNN - Ren 2015]



[NVIDIA dev blog]



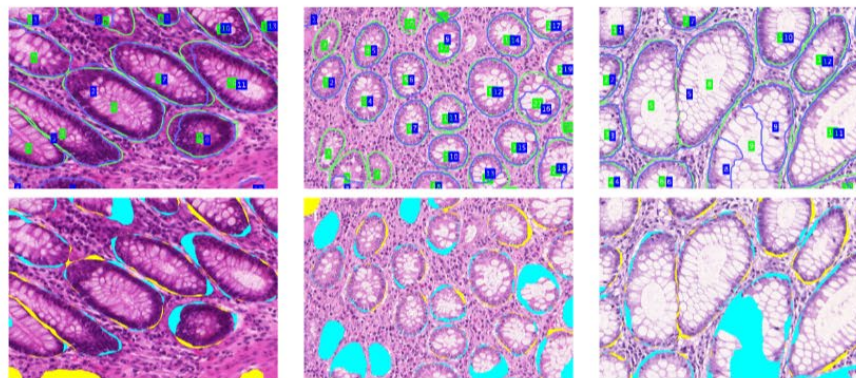
# What can Deep Learning do?



## Vision



[Stanford 2017]



(d) benign

(e) benign

(f) malignant

[Nvidia Dev Blog 2017]

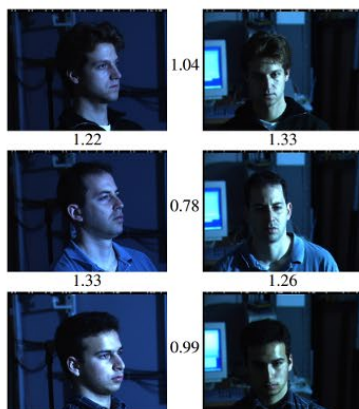
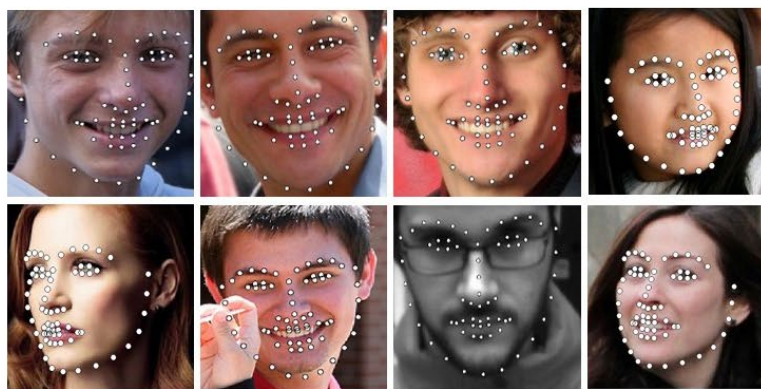


Figure 1. Illumination and Pose invariance.

[FaceNet - Google 2015]



[Facial landmark detection CUHK 2014]

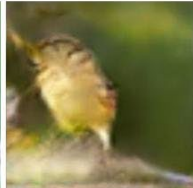
# What can Deep Learning do?



## Generative models for image



Sampled celebrities [Nvidia 2017]

| Text description | This bird is blue with white and has a very short beak                              | This bird has wings that are brown and has a yellow belly                           | A white bird with a black crown and yellow beak                                      | This bird is white, black, and brown in color, with a brown beak                      | The bird has small beak, with reddish brown crown and gray belly                      | This is a small, black bird with a white breast and white on the wingbars.            | This bird is white black and yellow in color, with a short black beak                 |
|------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Stage-I images   |   |   |   |   |   |   |   |
| Stage-II images  |  |  |  |  |  |  |  |

StackGAN v2 [Zhang 2017]



# What can Deep Learning do?



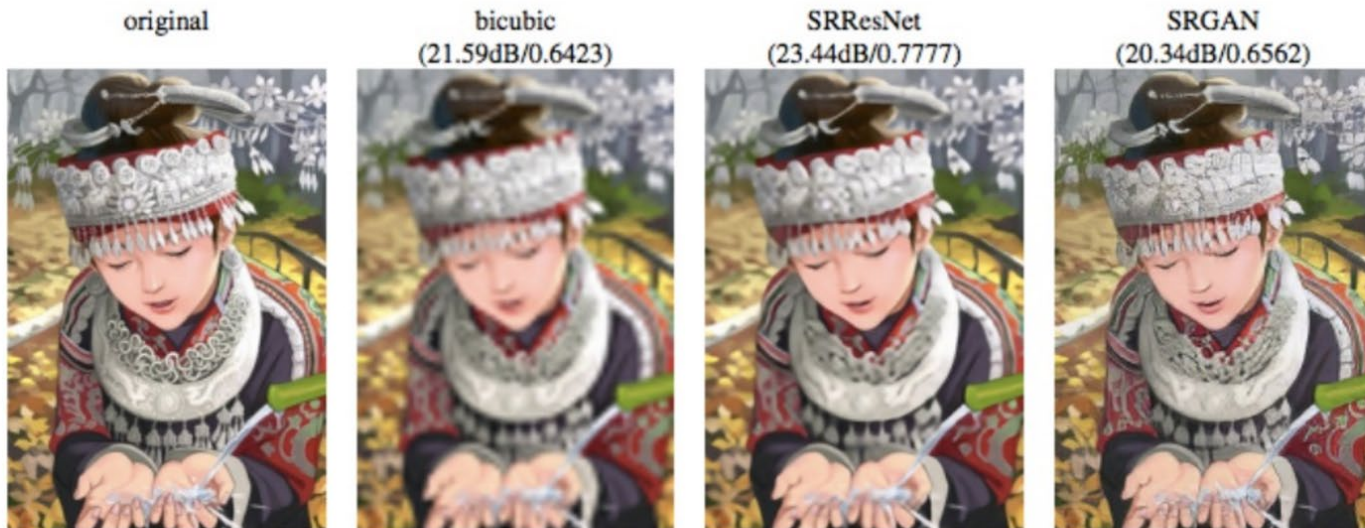
## Image Translation



[DeepDream 2015]



[Gatys 2015]



[Ledig 2016]

## Image Translation



Figure 4: Dog breed translation results.

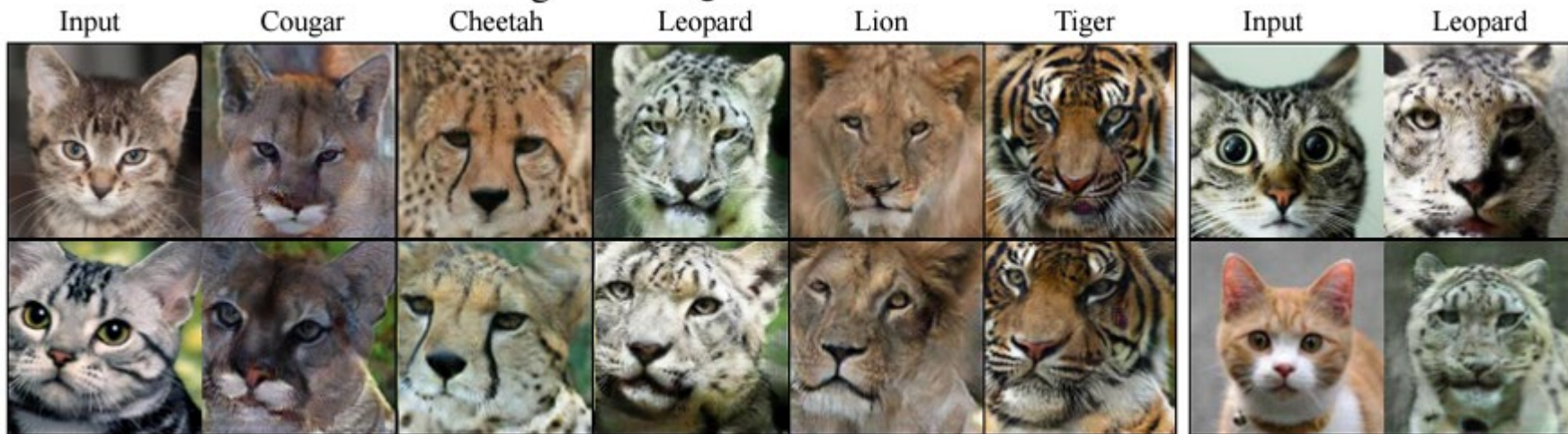


Figure 5: Cat species translation results.

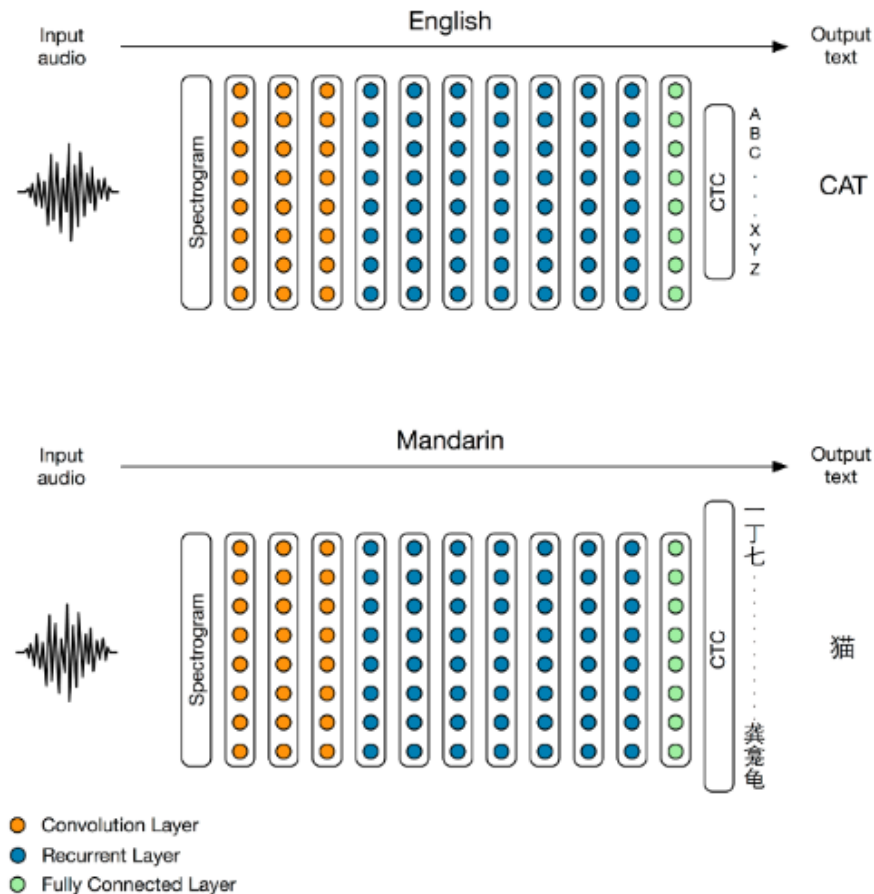
[Liu et al., 2017]



# What can Deep Learning do?



## Speech to Text



[Baidu 2014]



Hey Siri



# What can Deep Learning do?

## Natural Language Processing

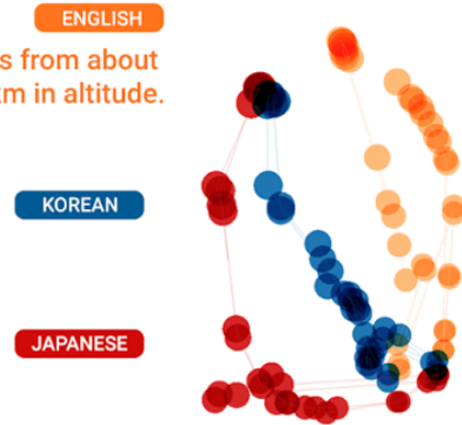
Translate

Italian Chinese French English - detected   
 deep learning   
 13/5000

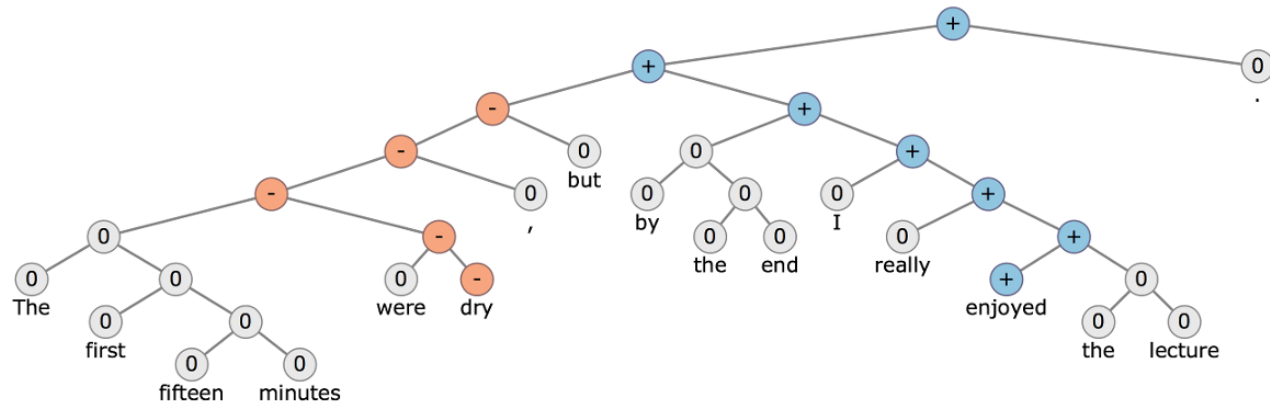
English Spanish French   
 l'apprentissage en profondeur   
 ☆ 📄 🔊 🔗

See also  
deep, learning

The stratosphere extends from about 10km to about 50km in altitude.



[Google Translate System - 2016]



[Socher 2015]

# What can Deep Learning do?



## Natural Language Processing



[Google Inbox Smart Reply]



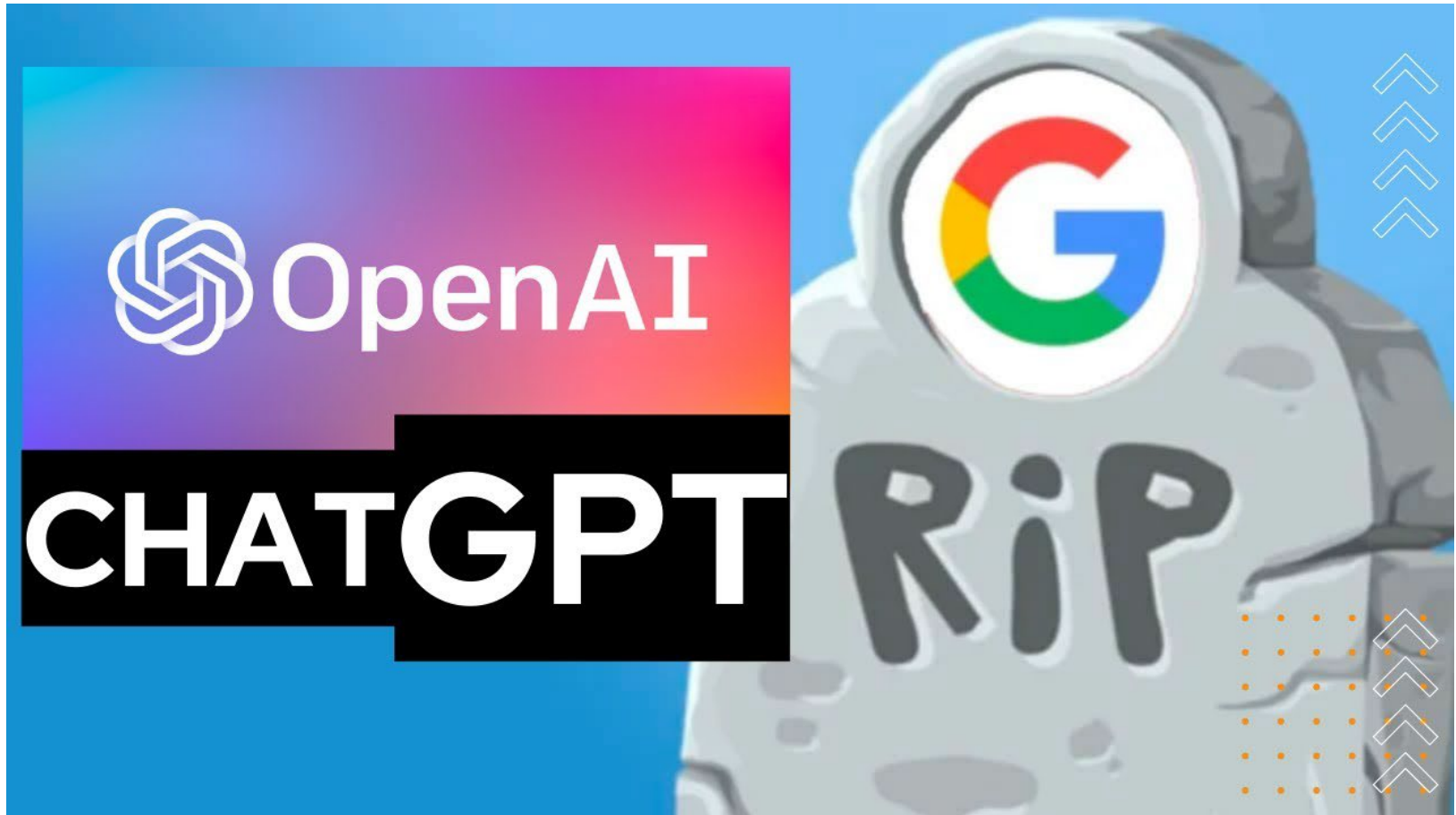
[Amazon Echo / Alexa]



# What can Deep Learning do?



## Natural Language Processing



**OpenAI ChatGPT is going to make Google Obsolete for Programmers**

Source: <https://www.youtube.com/watch?v=Ih1FxMGfUtU>

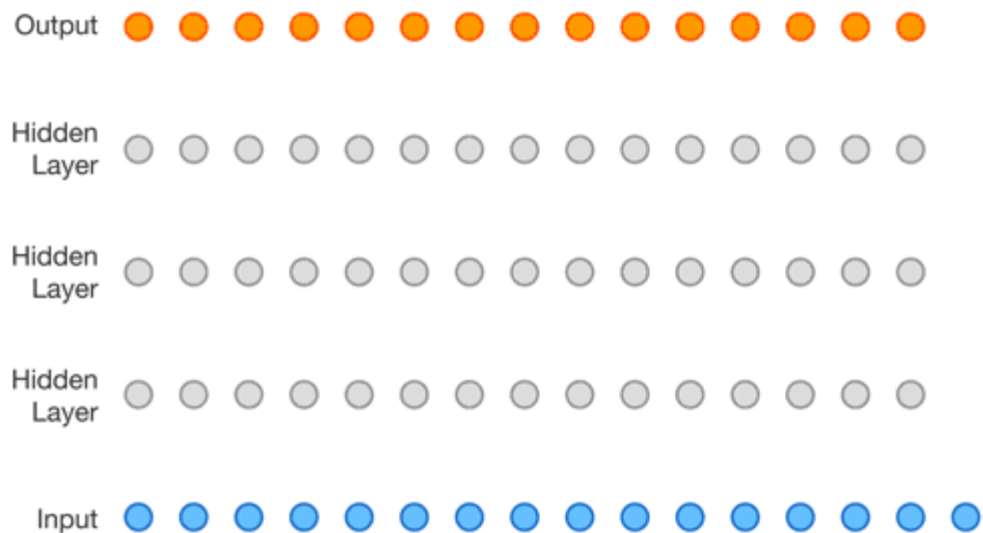
# What can Deep Learning do?



|                                                                                     |                                                       |                                                                                            |
|-------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------|
|    | GPT-4.1 (2025-04-14)<br>Text / Image Chat             |                                                                                            |
|    | GPT-4.1-mini (2025-04-14)<br>Text / Image Chat        |                                                                                            |
|    | GPT-5 (2025-08-07)<br>Text / Image Reasoning          |                                                                                            |
|    | GPT-5-mini (2025-08-07)<br>Text / Image Reasoning     |                                                                                            |
|    | Gemini 2.5 Pro (GA)<br>Text / Image Reasoning         |                                                                                            |
|    | Gemini 2.5 Flash (GA)<br>Text / Image Reasoning       |                                                                                            |
|    | Qwen3-235B-A22B-Thinking-2507<br>Text Reasoning       |  Free   |
|   | Qwen2.5-VL-72B-Instruct<br>Text / Image Chat          |  Free |
|  | DeepSeek-R1-0528 (Azure AI Foundry)<br>Text Reasoning |                                                                                            |
|  | DeepSeek-R1-0528 (Alibaba Cloud Bailian)              |                                                                                            |

Try: <https://genai.polyu.edu.hk/>

## Generative models for Audio



Sound generation with WaveNet [DeepMind 2017]

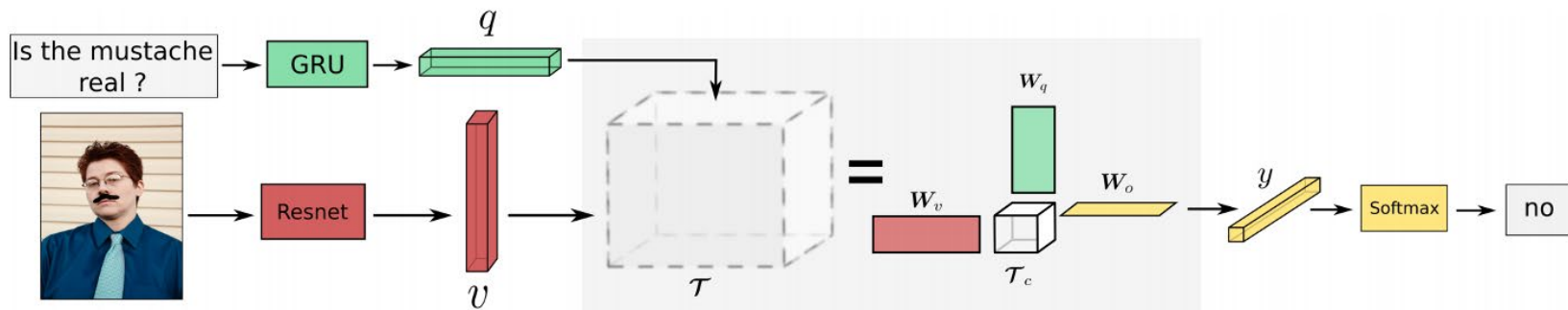
Which one do you think is generated?



*Tacotron 2 Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram*

# What can Deep Learning do?

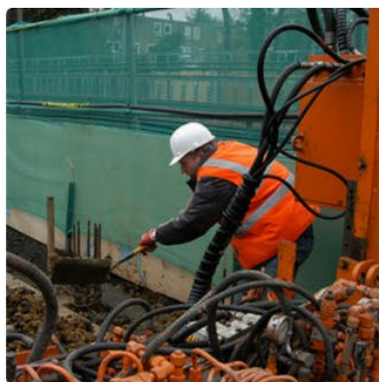
## Vision + NLP



[VQA - Mutan 2017]



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."



"boy is doing backflip on wakeboard."

[Karpathy 2015]

# What can Deep Learning do?



## Vision + NLP

TEXT PROMPT

an armchair in the shape of an avocado [...]

AI-GENERATED IMAGES



[View more or edit prompt ↓](#)

TEXT PROMPT

a store front that has the word 'openai' written on it [...]

AI-GENERATED IMAGES



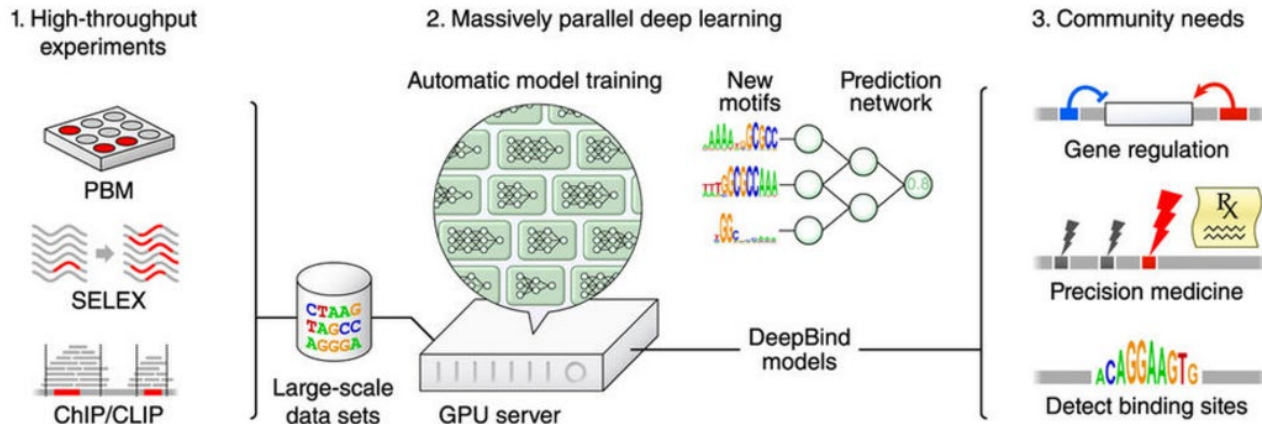
[View more or edit prompt ↓](#)

Open-AI GPT-3, or DALL-E: <https://openai.com/blog/dall-e/>

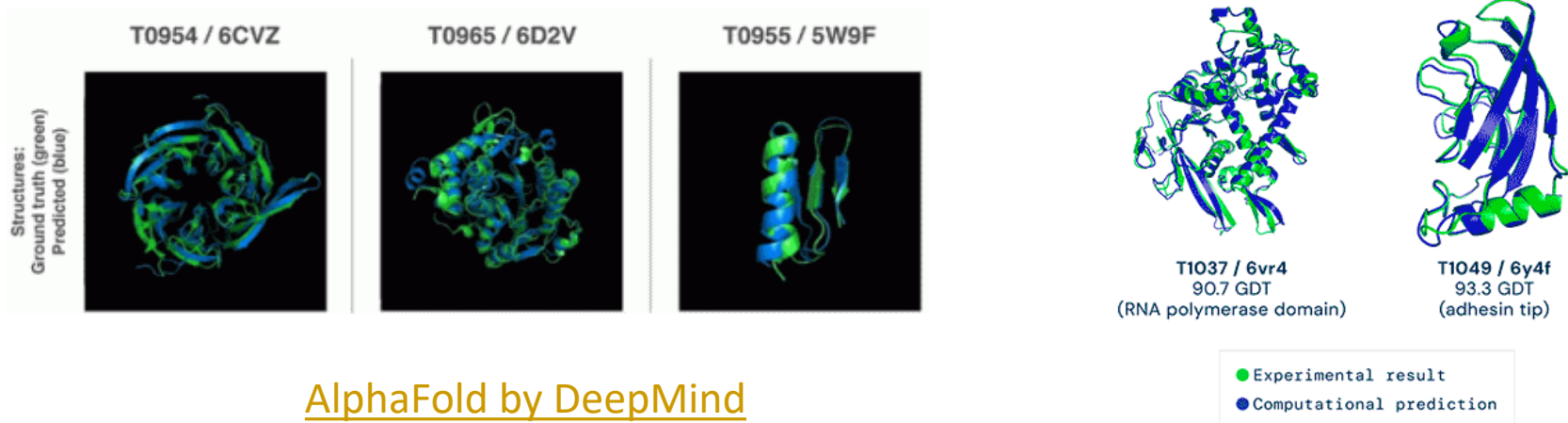


# What can Deep Learning do?

## Genomics



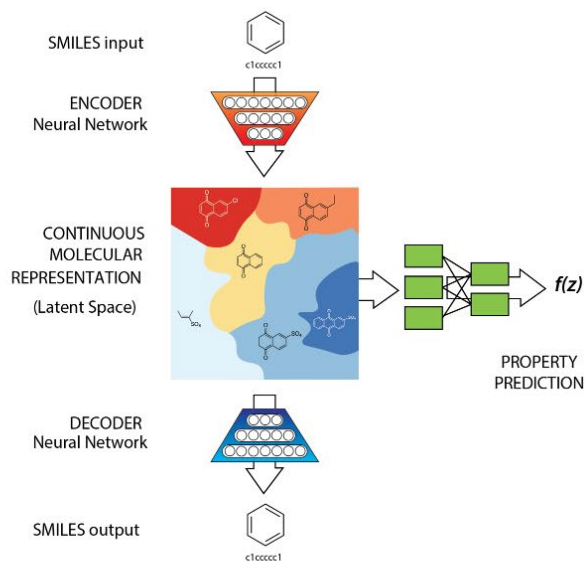
[Deep Genomics 2017]



AlphaFold by DeepMind



## Chemistry Physics



[Gómez-Bombarelli 2016]



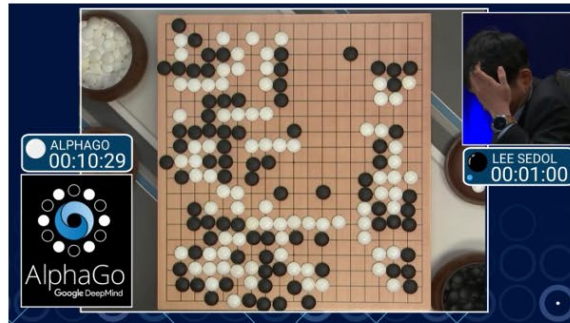
[Tompson 2016]



Finite element simulator accelerated (~100 fold) by a 3D convolutional network

# What can Deep Learning do?

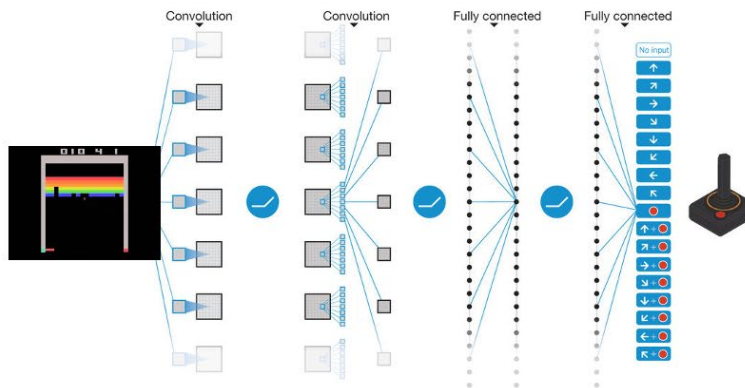
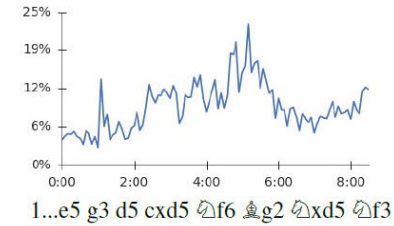
## Deep Learning for AI



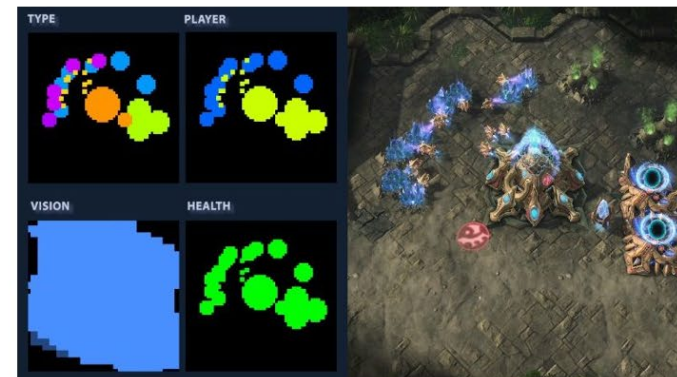
[Deepmind AlphaGo / Zero 2017]



A10: English Opening



[Atari Games - DeepMind 2016]



[Starcraft 2 for AI research]

AlphaGo/Zero: Monte Carlo Tree Search, Deep Reinforcement Learning, self-play

# What can Deep Learning do?



**Won IMO Gold Medal 2025**



# What can Deep Learning do?



## Image editing by Gemini



Prompt: remove the helmet.

# What can Deep Learning do?



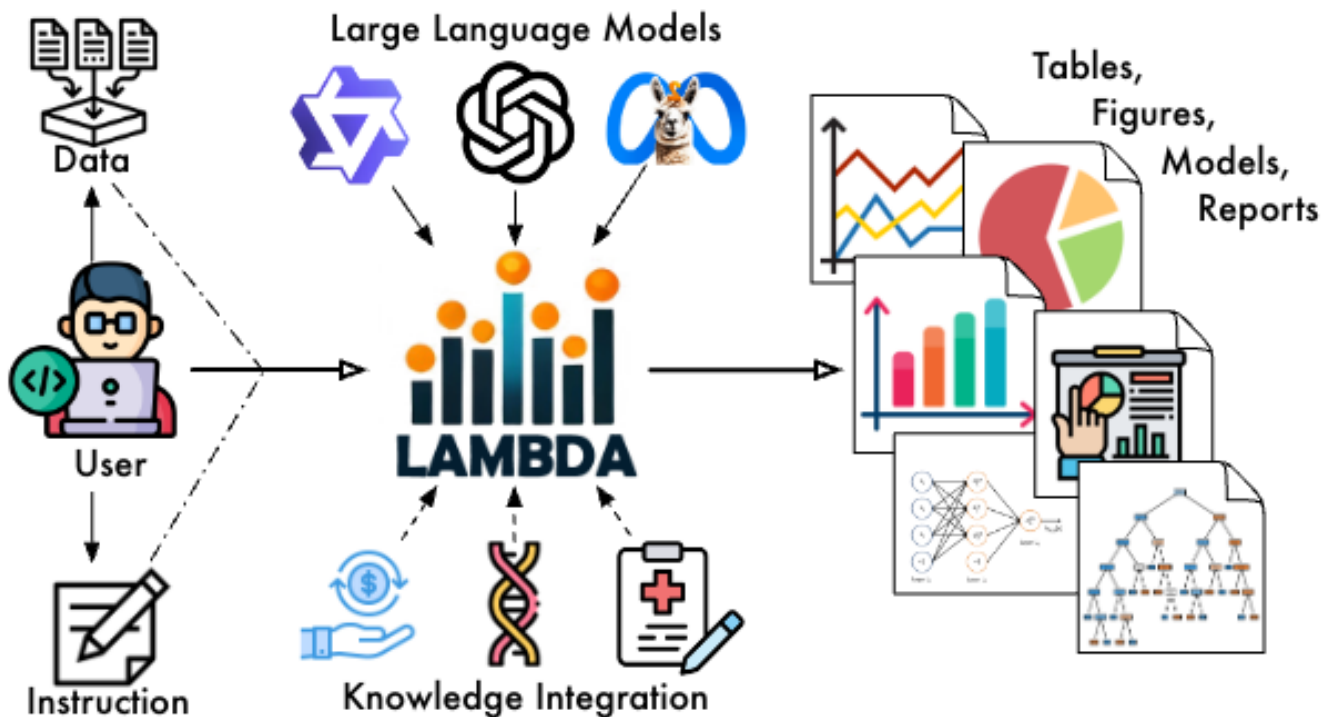
## Image editing by Gemini



Prompt: Change the head piece to something made from red flowers.

# What can Deep Learning do?

## LAMBDA: A Large Model Based Data Agent



<https://www.polyu.edu.hk/ama/cmfaai/lambda.html>



# **Deep Neural Networks**

**with**

## **motivating examples**

## An Example



- Least squares regression

- Data  $(X_i, Y_i), i = 1, \dots, n$ .

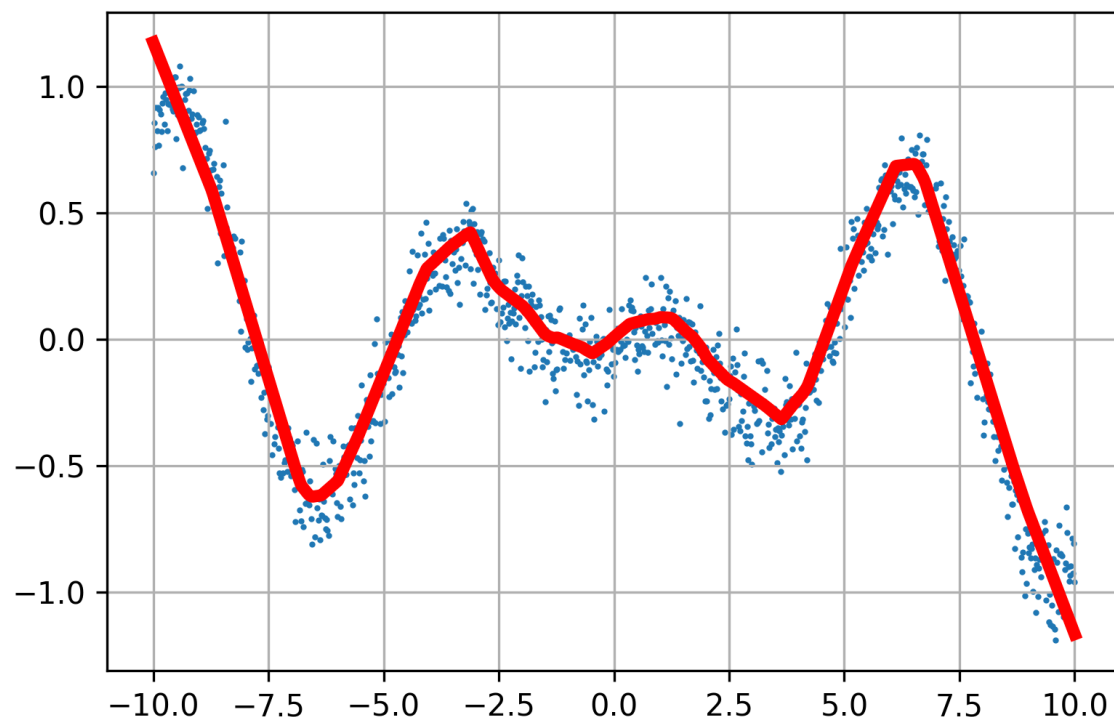
- To find a  $f(x; \alpha, \beta) = \beta'x + \alpha$  such that

$\sum (Y_i - f(X_i))^2$   
is minimized over

$$\mathcal{F} = \{f: f(x; \alpha, \beta) = \beta'x + \alpha, \alpha, \beta \in \mathbb{R}\}$$

- The model  $f(x; \alpha, \beta)$  is **linear** in the input  $x$ .
- Simple, efficient, easy to compute.
- Not **expressive**, especially for nonlinear data.

## An Example



- Data  $(X_i, Y_i), i = 1, \dots, n$ .

- To find a network

$$f(x; \theta)$$

such that

$\sum (Y_i - f(X_i))^2$   
is minimized over

$\mathcal{F} = \{f: f(x; \theta) \text{ is a}$   
*neural network*  
*parameterized by  $\theta \in \mathbb{R}^s$*  }

- The model  $f(x; \theta)$  is a **neural network** parameterized by  $\theta$
- Can have **many parameters**, need training.
- Very **expressive**, especially for **high-dimensional nonlinear** data.

# **Mathematical Definition of Deep Neural Networks**

## Feedforward Neural Networks (Multi-Layer Perceptrons)

- The architecture of a MLP is expressed as a composition of a series of functions

$$f_{\theta}(x) = \mathcal{A}_L \circ \sigma \circ \mathcal{A}_{L-1} \circ \sigma \circ \dots \circ \sigma \circ \mathcal{A}_1(x), \quad x \in \mathbb{R}^{d_0}$$

where

$$\mathcal{A}_i(x) = W_i x + b_i, \quad x \in \mathbb{R}^{d_{i-1}}$$

is the  $i$ -th linear transformation with

weight matrix  $W_i \in \mathbb{R}^{d_i \times d_{i-1}}$  and bias vector  $b_i \in \mathbb{R}^{d_i}$ ,

and  $\sigma$  is the activation function,

e.g., ReLU  $\sigma(x) = \max\{x, 0\}$ . Sigmoid  $\sigma(x) = (1 + e^{-x})^{-1}$ .



## Feedforward Neural Networks (Multi-Layer Perceptrons)

- The architecture of a MLP is expressed as a **composition of a series of functions**

$$f_{\theta}(x) = \mathcal{A}_L \circ \sigma \circ \mathcal{A}_{L-1} \circ \sigma \circ \dots \circ \sigma \circ \mathcal{A}_1(x), \quad x \in \mathbb{R}^{d_0}$$

where

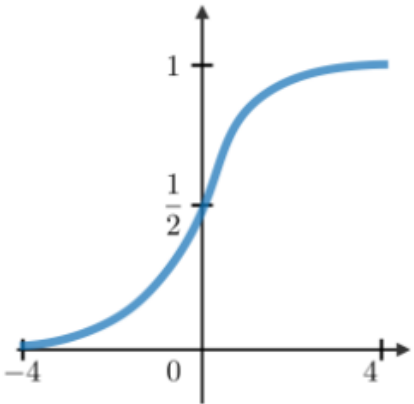
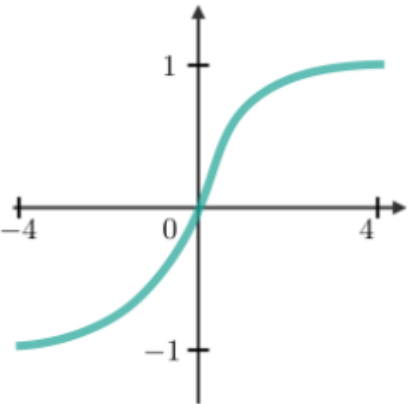
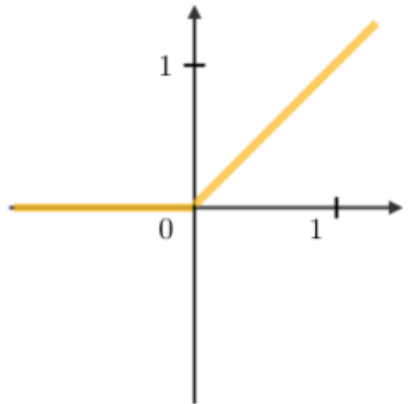
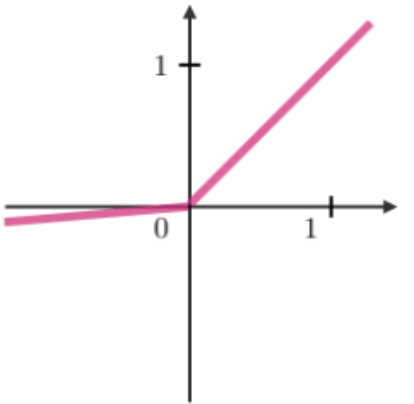
$$\mathcal{A}_i(x) = W_i x + b_i, \quad x \in \mathbb{R}^{d_{i-1}}.$$

- Such MLP has  **$L$  layers**,  
The **input  $x$**  is the **0-th layer** and the output is the last layer.  
 $h_i(x) = \sigma \circ \mathcal{A}_i \circ \sigma \circ \dots \circ \sigma \circ \mathcal{A}_1(x)$  is the  $i$ -th layer, **hidden layer**.  
The **depth of the network** ( $L - 1$ ), number of hidden layers.
- The **width** of the  $i$ -th layer is  $d_i$ . The **width of the network** is  $\max\{d_1, \dots, d_{L-1}\}$ .
- $\theta = (W_1, b_1, \dots, W_L, b_L)$  collects all the parameters. **Size of the network** is  $\# \theta$ .



## Activation functions

$$f_{\theta}(x) = \mathcal{A}_L \circ \sigma \circ \mathcal{A}_{L-1} \circ \sigma \circ \dots \circ \sigma \circ \mathcal{A}_1(x), \quad x \in \mathbb{R}^{d_0}$$

| Sigmoid                                                                           | Tanh                                                                               | ReLU                                                                                 | Leaky ReLU                                                                           |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $g(z) = \frac{1}{1 + e^{-z}}$                                                     | $g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$                                         | $g(z) = \max(0, z)$                                                                  | $g(z) = \max(\epsilon z, z)$<br>with $\epsilon \ll 1$                                |
|  |  |  |  |

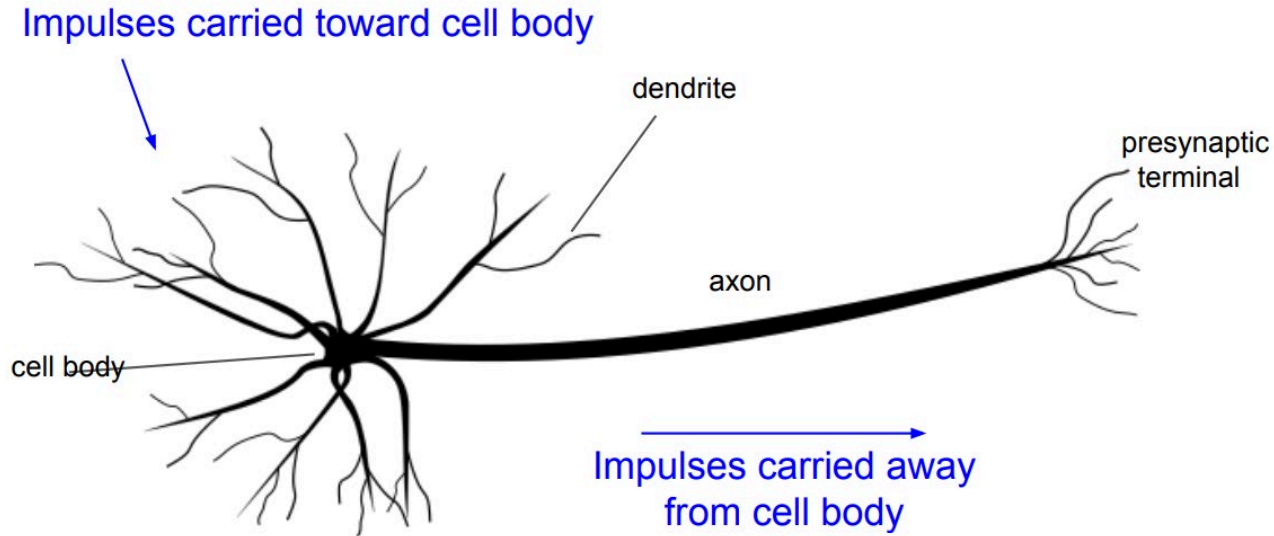
An Activation functions is considered non-saturated if  $\lim_{x \rightarrow \infty} \sigma(x) = \infty$ .

**Saturated Nonlinearity** : Sigmoid, Tanh

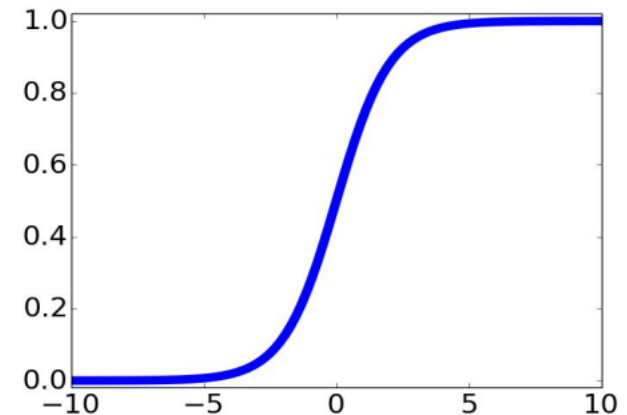
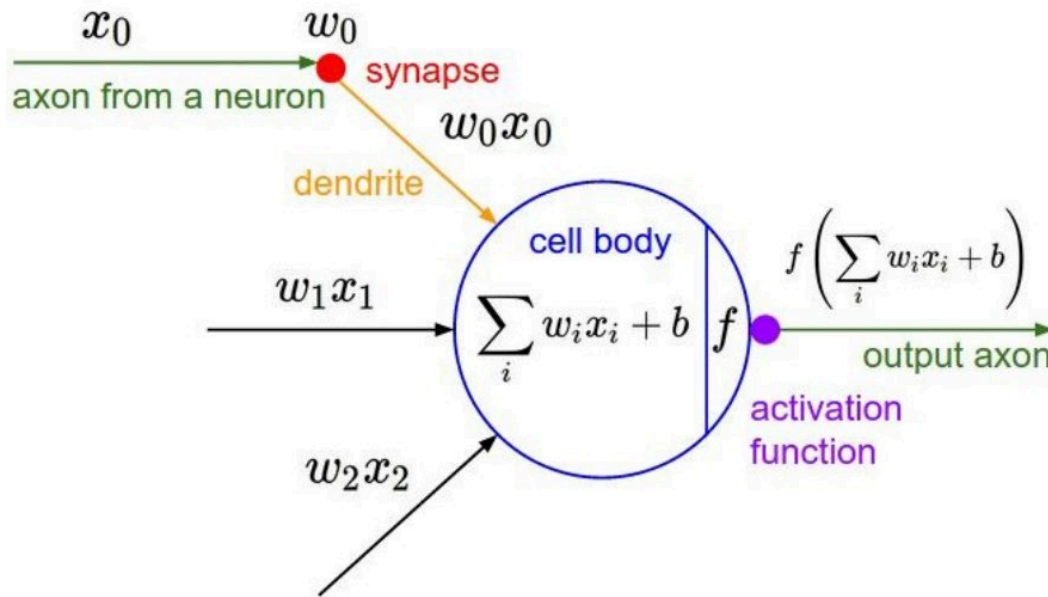
**Non-saturated Nonlinearity**: Rectified Linear Unit (ReLU), Leaky ReLU

ReLU is a good default choice for most problems.

## Biological Neurons & Neurons in a Network

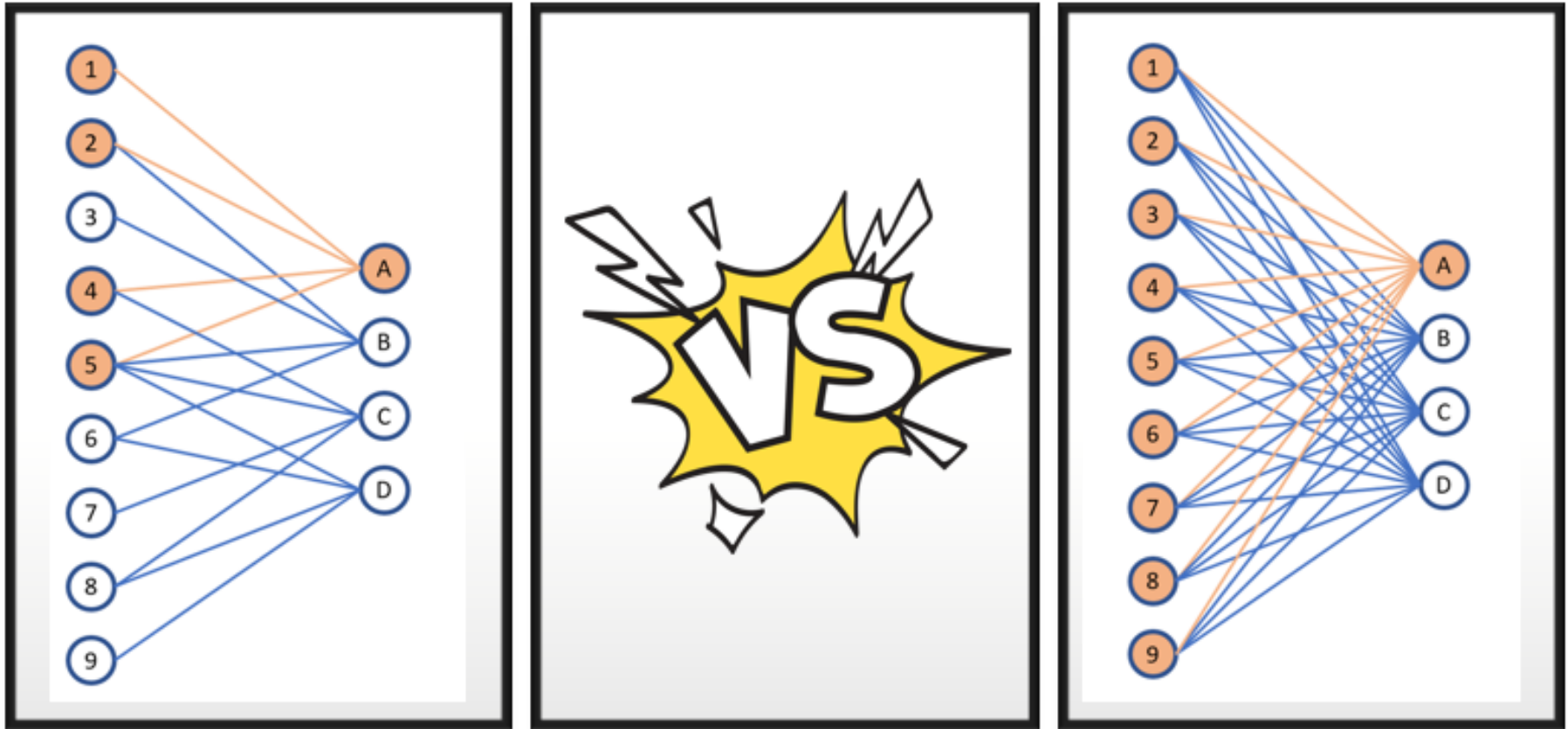


[This image](#) by Felipe Perucho is licensed under CC-BY 3.0.



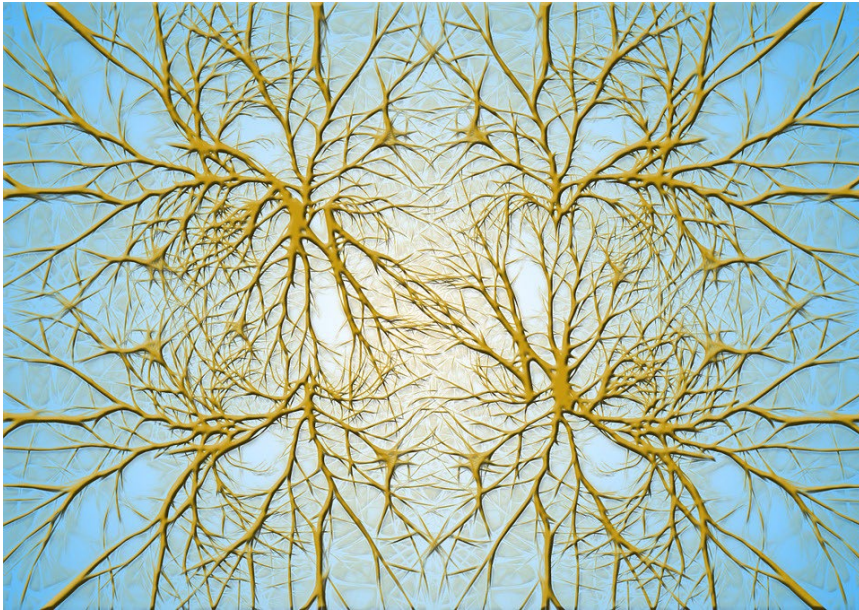
Sigmoid activation function  $\frac{1}{1+e^{-x}}$

## Feedforward Neural Networks (Multi-Layer Perceptrons)



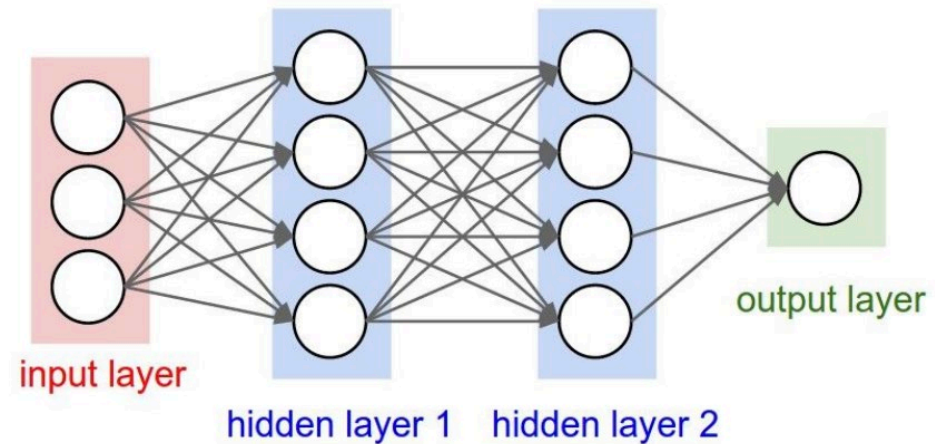
**Layers in MLP are fully connected**

## Biological Neurons



Complex connectivity

## Neurons in a neural network:



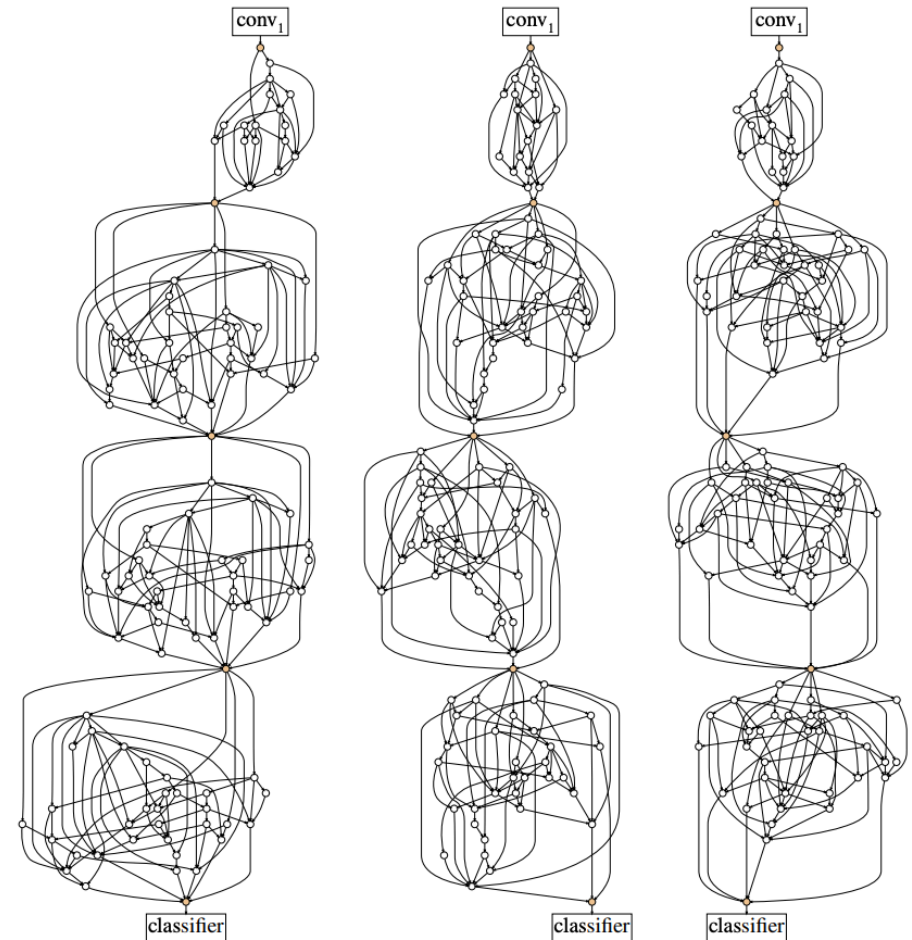
Layer structure  
Fully connected  
Computational efficient

## Randomly wired neural networks

generated by the classical Watts-Strogatz (WS) model: these three instances of random networks achieve (left-to-right)

79.1%, 79.1%, 79.0% classification accuracy on ImageNet under a similar computational budget to ResNet-50, which has 77.1% accuracy.

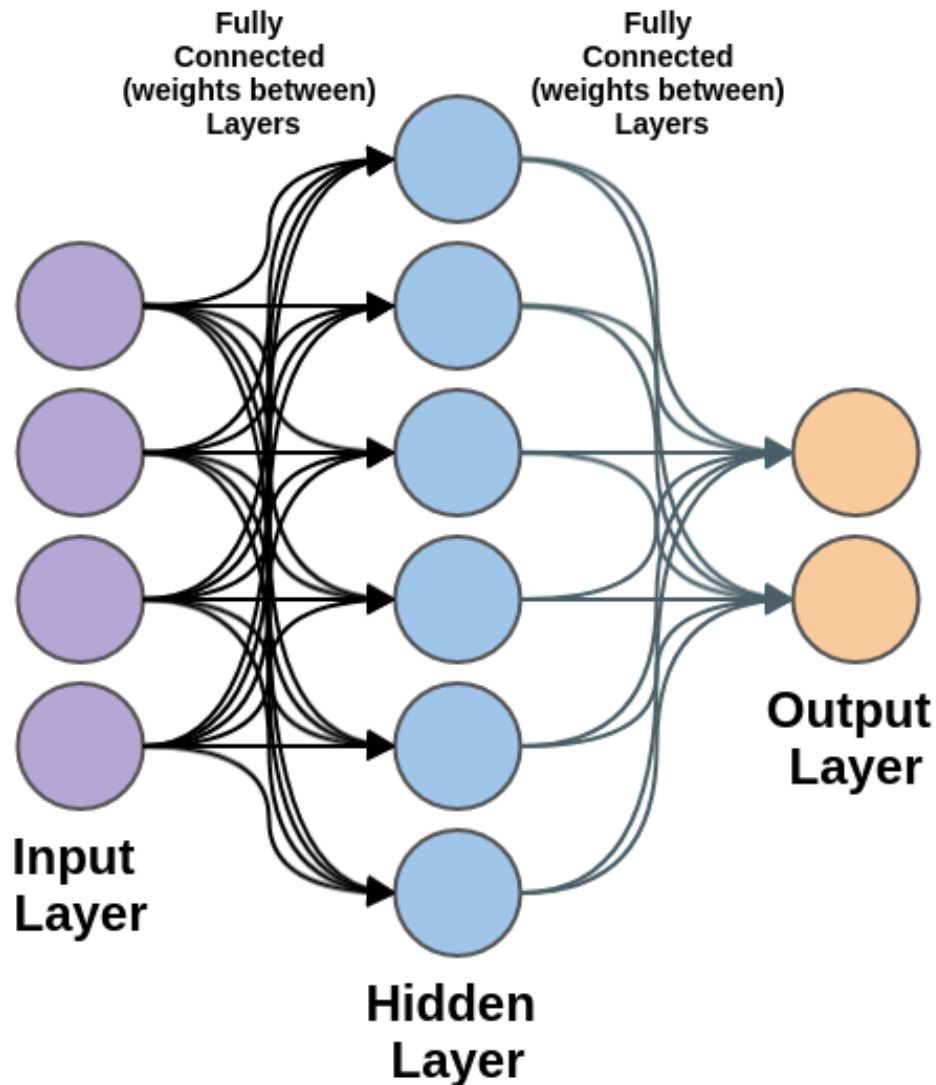
## Randomly wired neural networks can also work!



Xie S, Kirillov A, Girshick R, He K. Exploring randomly wired neural networks for image recognition. In Proceedings of the IEEE/CVF International Conference on Computer Vision 2019 (pp. 1284-1293).



## Examples of Multi-Layer Perceptron (structure)



- 2-layer MLP with **1** hidden layer, **depth** = 1
- Network **width** = 6
- **Size** =  $(4 \cdot 6 + 6) + (6 \cdot 2 + 2) = 44$
- $f_{\theta}(x) = W_2 \sigma(W_1 x + b_1) + b_2$

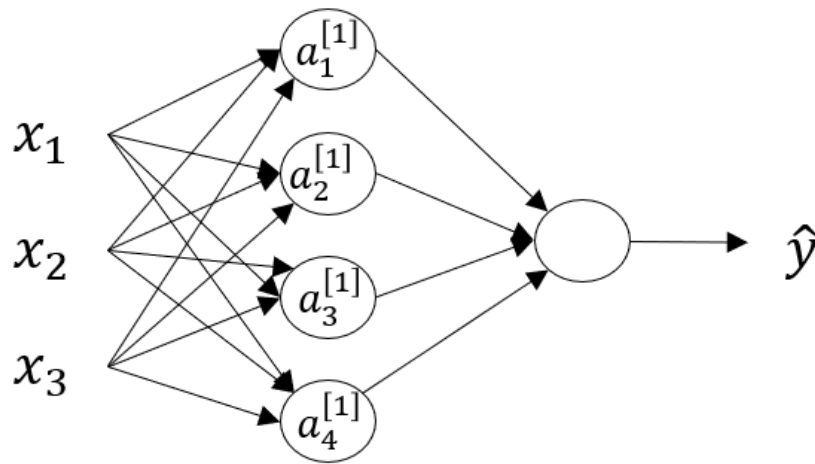
where  $x \in \mathbb{R}^4$ ,

$$W_1 \in \mathbb{R}^{6 \times 4}, b_1 \in \mathbb{R}^6,$$

$$W_2 \in \mathbb{R}^{2 \times 6}, b_2 \in \mathbb{R}^2.$$



## Examples of Multi-Layer Perceptron (computation)



$$z_1^{[1]} = w_1^{[1]T} x + b_1^{[1]}, \quad a_1^{[1]} = \sigma(z_1^{[1]})$$

$$z_2^{[1]} = w_2^{[1]T} x + b_2^{[1]}, \quad a_2^{[1]} = \sigma(z_2^{[1]})$$

$$z_3^{[1]} = w_3^{[1]T} x + b_3^{[1]}, \quad a_3^{[1]} = \sigma(z_3^{[1]})$$

$$z_4^{[1]} = w_4^{[1]T} x + b_4^{[1]}, \quad a_4^{[1]} = \sigma(z_4^{[1]})$$

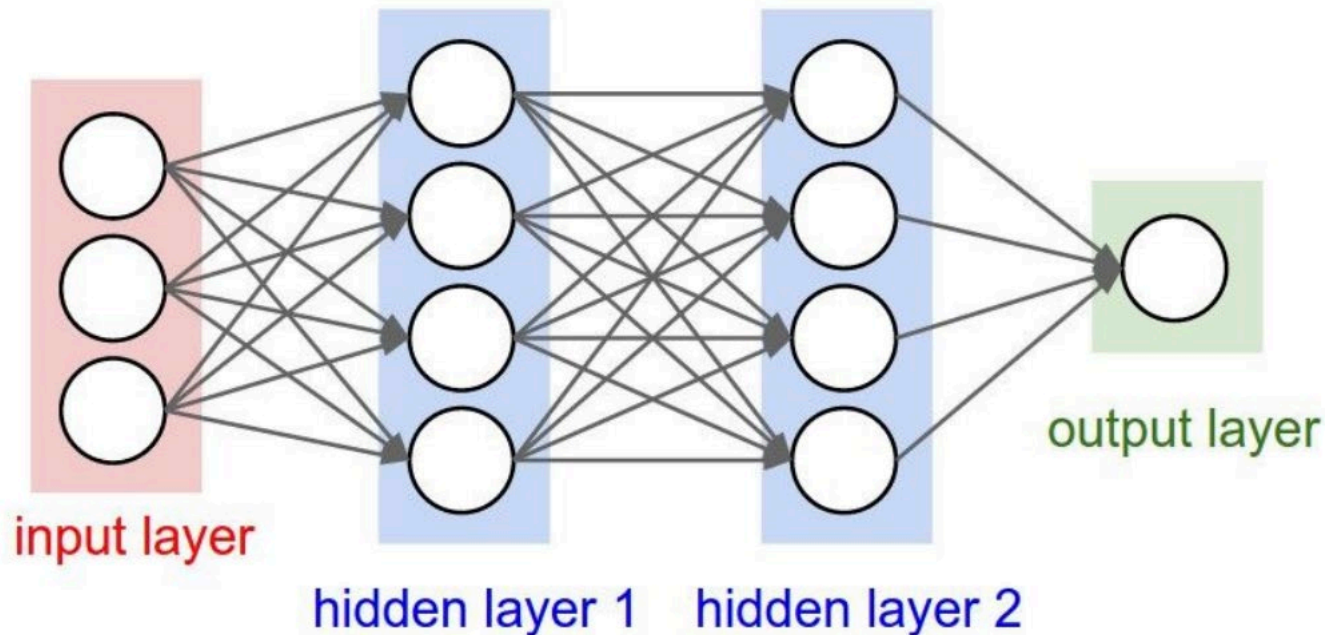
$$\hat{y} = w_1^{[2]} \cdot a_1^{[1]} + w_2^{[2]} \cdot a_2^{[1]} + w_3^{[2]} \cdot a_3^{[1]} + w_4^{[2]} \cdot a_4^{[1]} + b_1^{[2]}$$

**Different values of parameters (weights and bias) give different neural work.**

$$\text{Weights: } \mathbf{W}_1 = (w_1^{[1]}, w_2^{[1]}, w_3^{[1]}, w_4^{[1]}) \quad \mathbf{W}_2 = (w_1^{[2]}, w_2^{[2]}, w_3^{[2]}, w_4^{[2]})$$

$$\text{Biases: } \mathbf{b}_1 = (b_1^{[1]}, b_2^{[1]}, b_3^{[1]}, b_4^{[1]}) \quad \mathbf{b}_2 = b_1^{[2]}$$

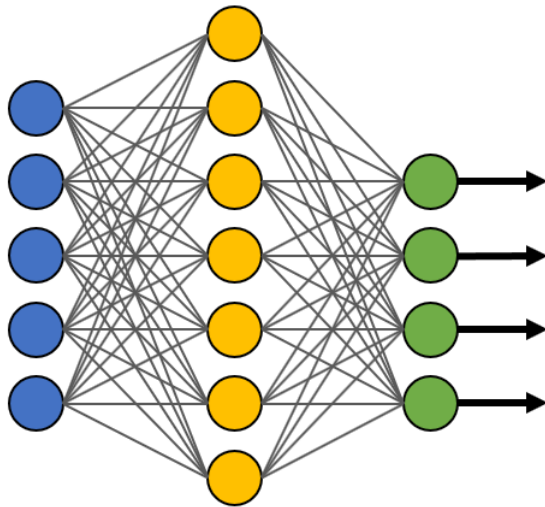
## Examples of Multi-Layer Perceptron (programming)



```
# forward-pass of a 3-layer neural network:  
f = lambda x: 1.0/(1.0 + np.exp(-x)) # activation function (use sigmoid)  
x = np.random.randn(3, 1) # random input vector of three numbers (3x1)  
h1 = f(np.dot(W1, x) + b1) # calculate first hidden layer activations (4x1)  
h2 = f(np.dot(W2, h1) + b2) # calculate second hidden layer activations (4x1)  
out = np.dot(W3, h2) + b3 # output neuron (1x1)
```

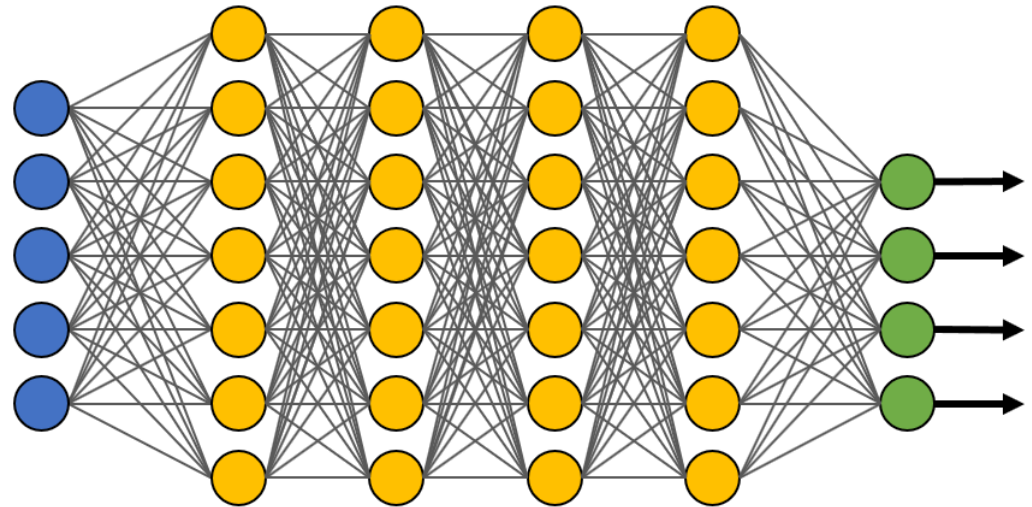
## From shallow to deep

### Shallow Neural Network



● Input Layer

### Deep Neural Networks

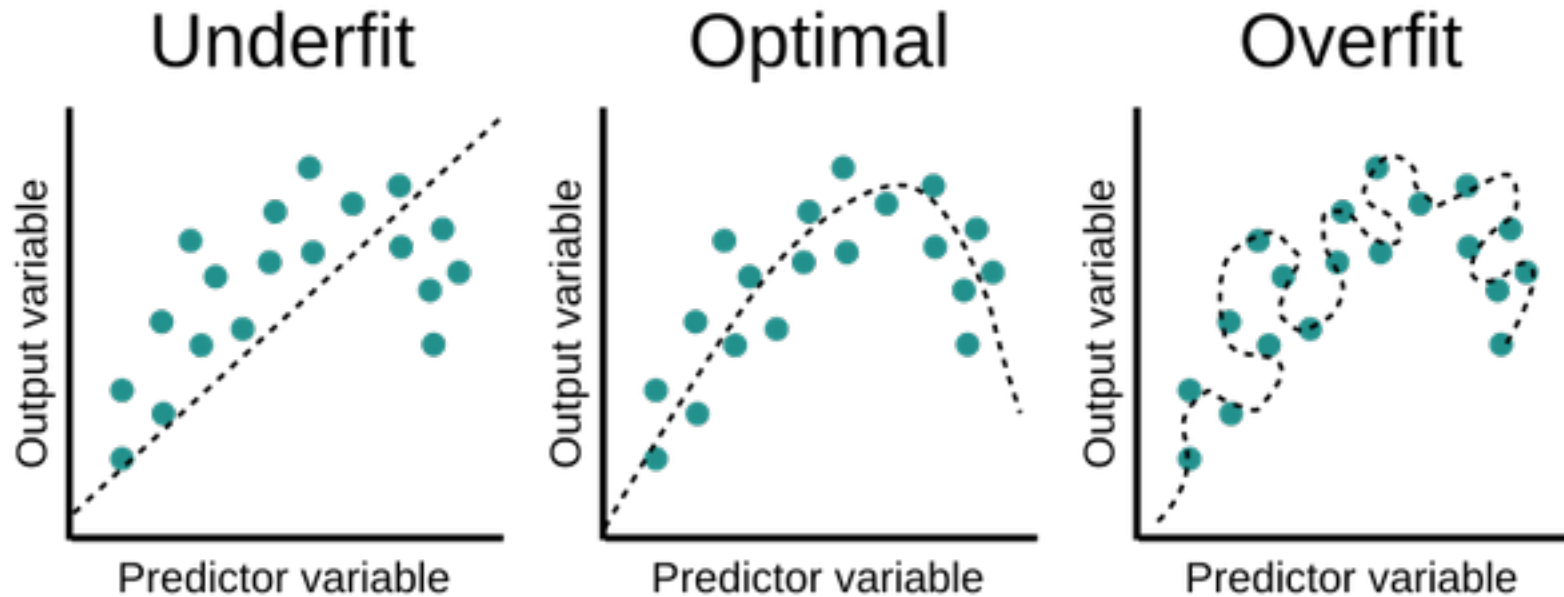


● Hidden Layer

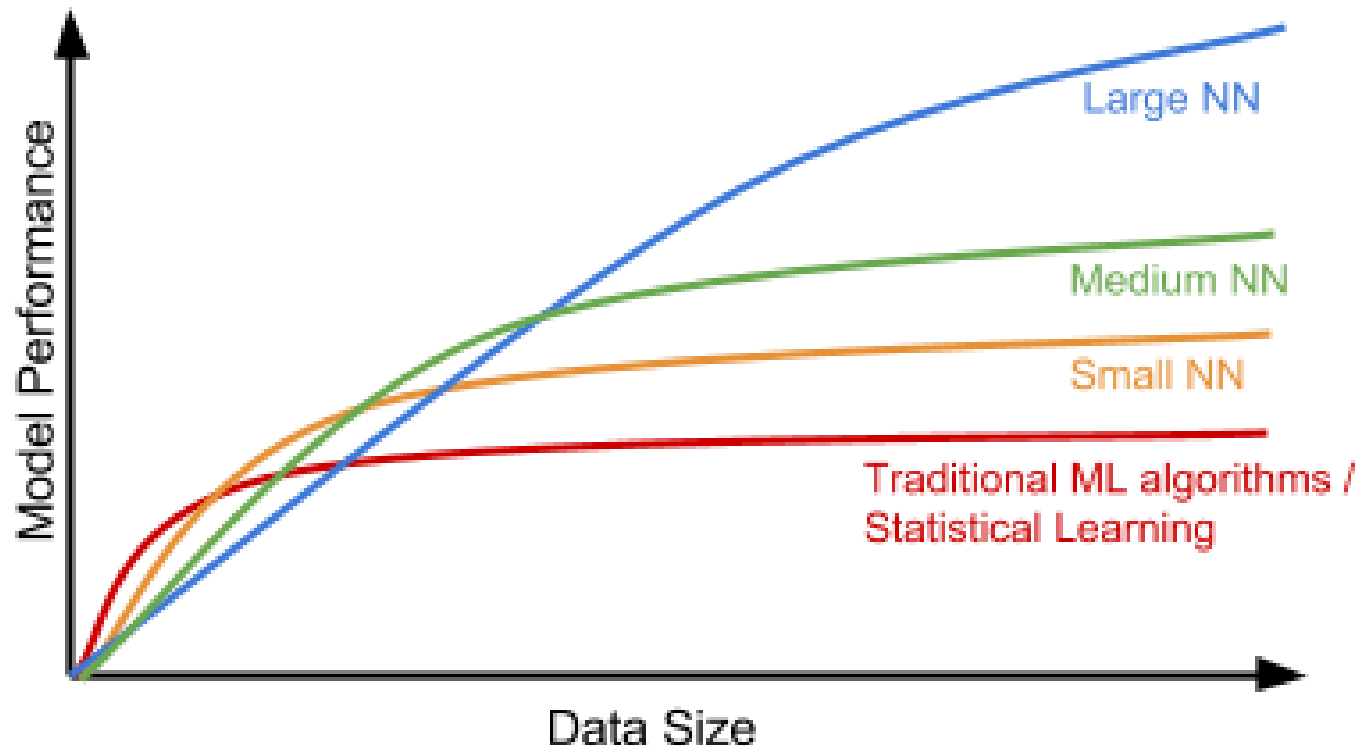
● Output Layer

From shallow to deep.

Do large Neural Networks overfit the data?



**Larger Neural Networks perform better on data with larger size**



Source: <https://lilianweng.github.io/posts/2017-06-21-overview/>

**Neural networks are composited functions.**

**Neural networks can approximate other functions.**

**Neural networks can be expressive.**





## Universality of Neural Networks (Arbitrary-width case)

**Universal approximation theorem:** Let  $C(X, Y)$  denote the set of **continuous functions** from  $X$  to  $Y$ . Let  $\sigma \in C(\mathbb{R}, \mathbb{R})$ . Note that  $(\sigma \circ x)_i = \sigma(x_i)$ , so  $\sigma \circ x$  denotes  $\sigma$  applied to each component of  $x$ .

Then  $\sigma$  is not **polynomial if and only if** for every  $n \in \mathbb{N}$ ,  $m \in \mathbb{N}$ , **compact**  $K \subseteq \mathbb{R}^n$ ,  $f \in C(K, \mathbb{R}^m)$ ,  $\varepsilon > 0$  there exist  $k \in \mathbb{N}$ ,  $A \in \mathbb{R}^{k \times n}$ ,  $b \in \mathbb{R}^k$ ,  $C \in \mathbb{R}^{m \times k}$  such that

$$\sup_{x \in K} \|f(x) - g(x)\| < \varepsilon$$

where

$$g(x) = C \cdot (\sigma \circ (A \cdot x + b))$$

**Any continuous functions** defined on a compact set can be approximated **arbitrarily well** by a **shallow neural network** if the shallow neural network is **arbitrarily wide**.

## Universality of Neural Networks (Arbitrary-depth case)

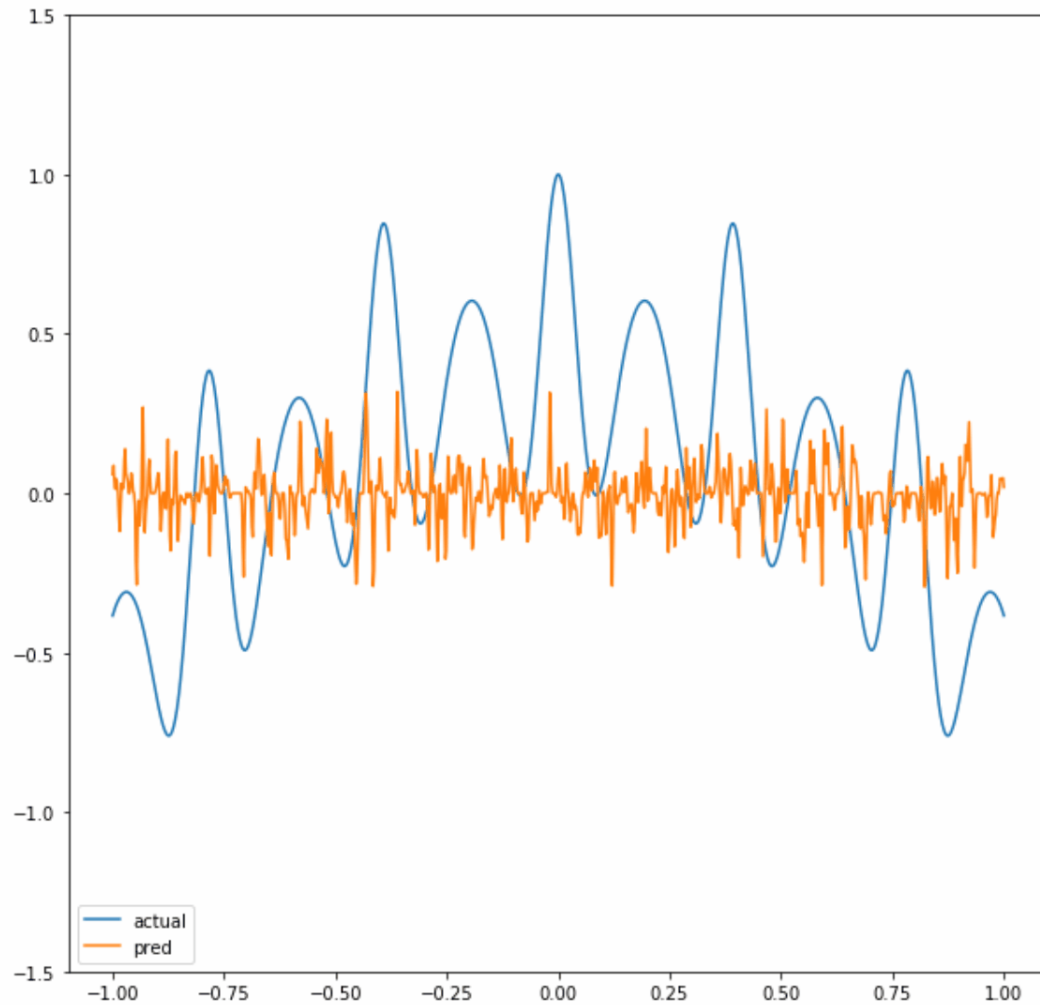
Universal approximation theorem (*L1 distance, ReLU activation, arbitrary depth, minimal width*). For any Bochner–Lebesgue  $p$ -integrable function  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  and any  $\epsilon > 0$ , there exists a fully-connected ReLU network  $F$  of width exactly  $d_m = \max\{n + 1, m\}$ , satisfying

$$\int_{\mathbb{R}^n} \|f(x) - F(x)\|^p dx < \epsilon.$$

Moreover, there exists a function  $f \in L^p(\mathbb{R}^n, \mathbb{R}^m)$  and some  $\epsilon > 0$ , for which there is no fully-connected ReLU network of width less than  $d_m = \max\{n + 1, m\}$  satisfying the above approximation bound.

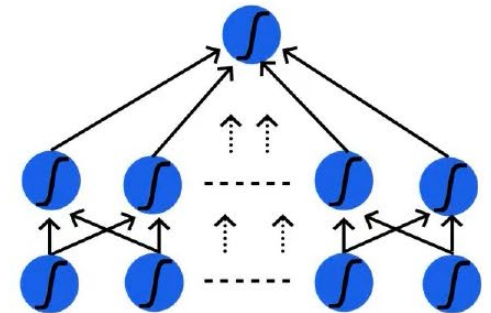
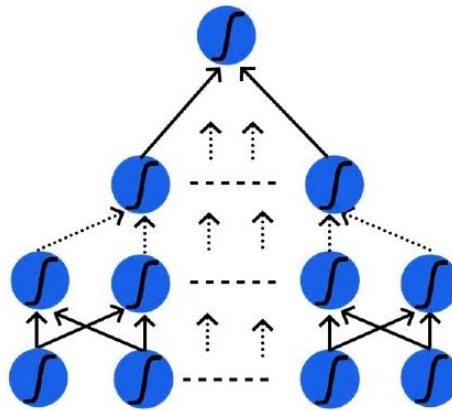
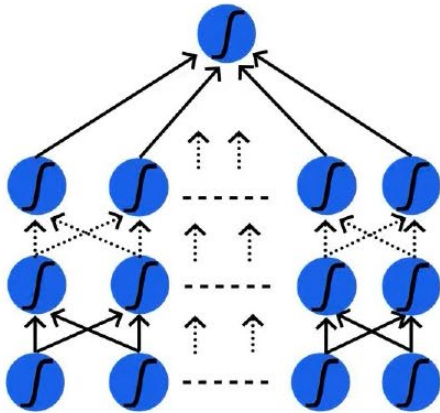
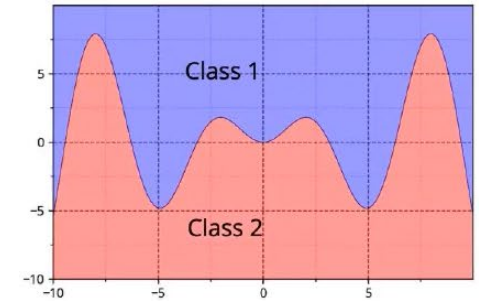
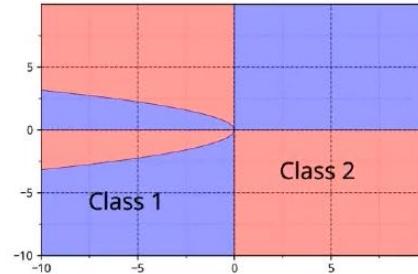
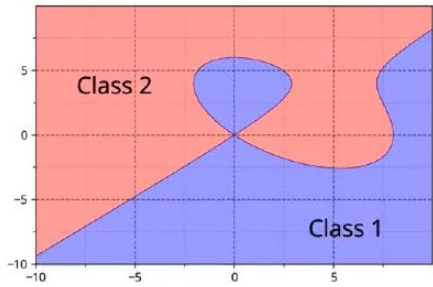
Any **continuous functions** defined on a compact set can be approximated **arbitrarily well** by a **fixed-width** neural network if the neural network is **arbitrarily deep**.

## Universality of Neural Networks

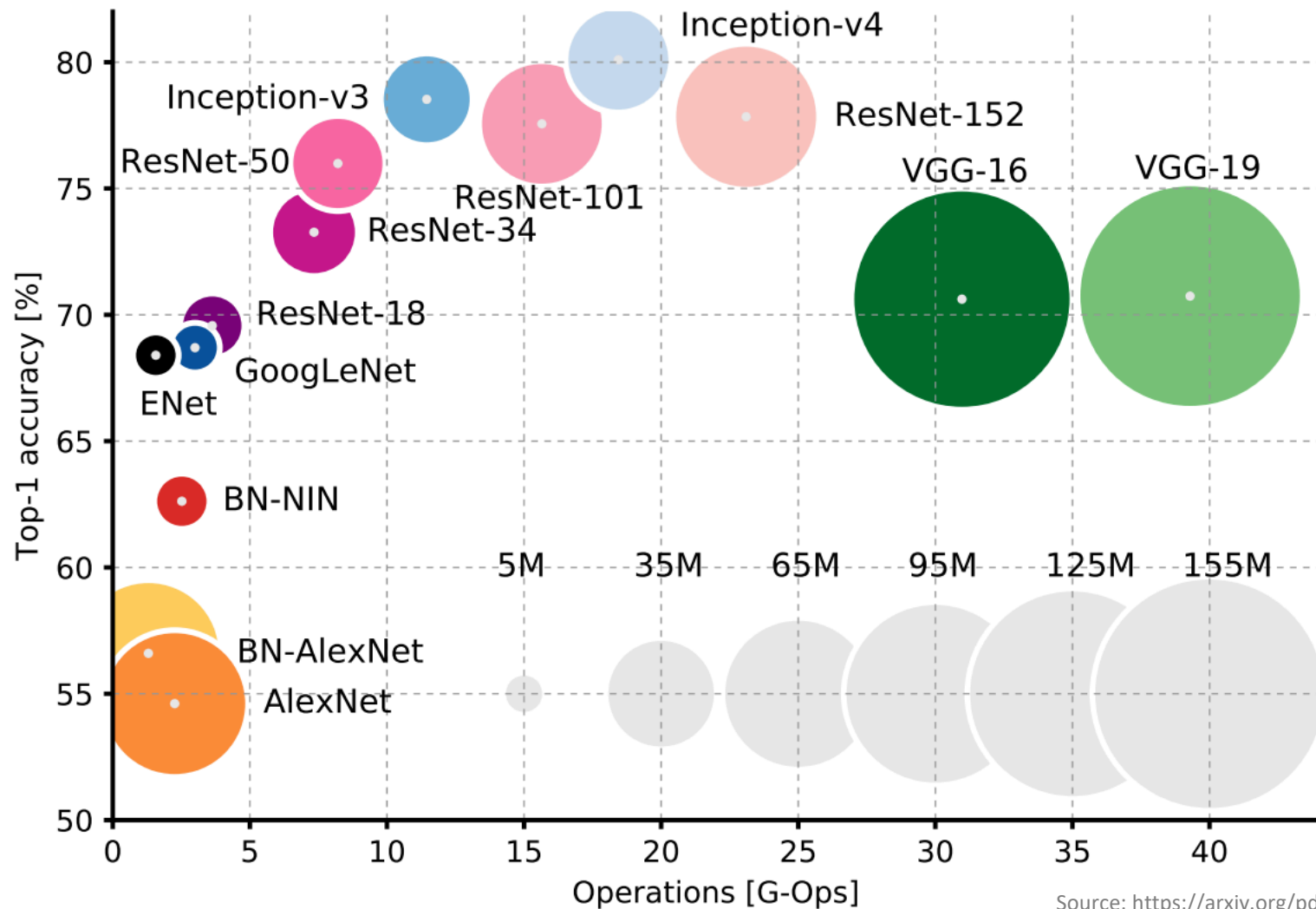


Neural networks can approximate regression functions very well.

## Universality of Neural Networks

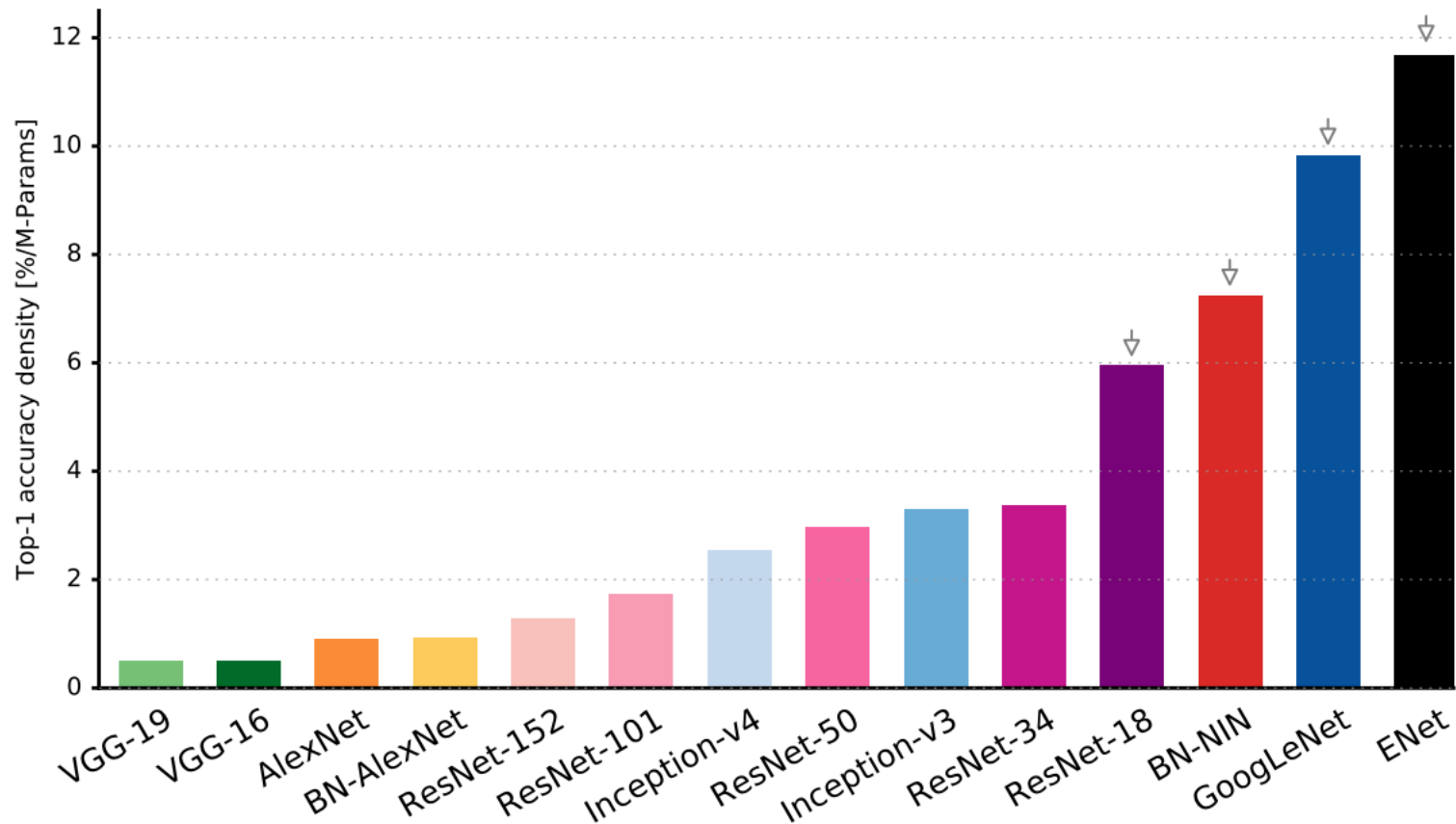


Neural networks can also approximate classification regions very well.



Source: <https://arxiv.org/pdf/1605.07678.pdf>

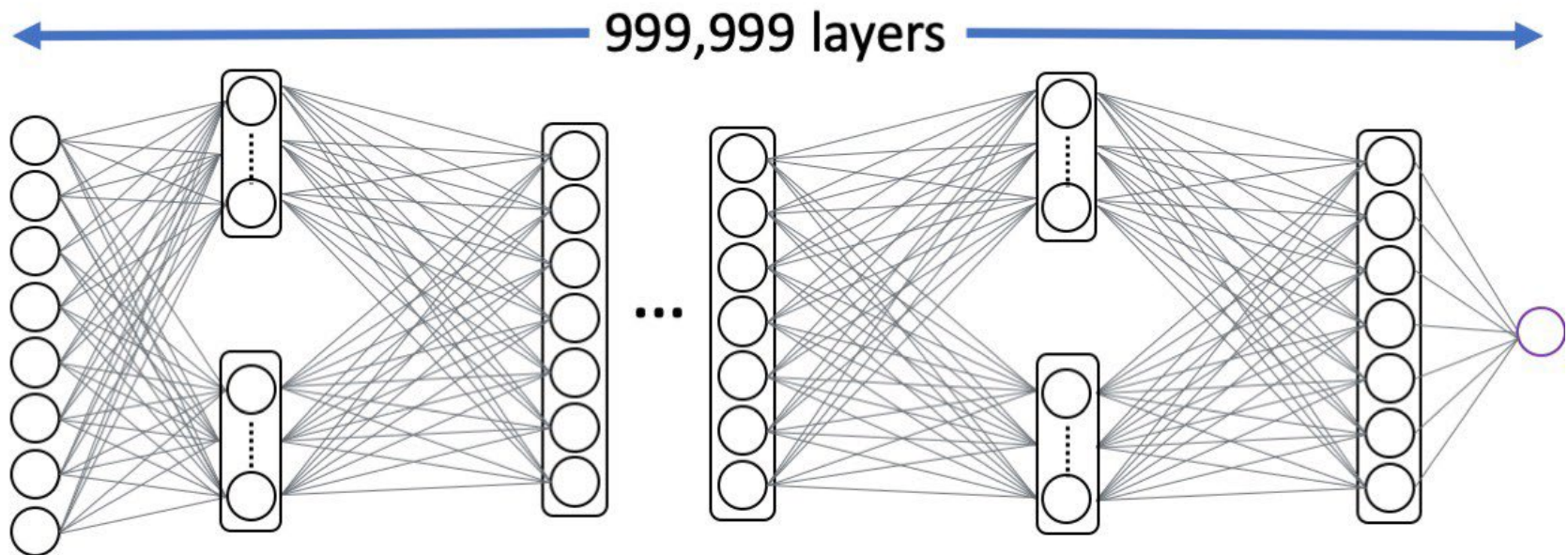
**Accuracy vs. operations, size  $\propto$  parameters.** Top-1 one-crop accuracy versus amount of operations required for a single forward pass. The size of the blobs is proportional to the number of network parameters; a legend is reported in the bottom right corner, spanning from  $5 \times 10^6$  to  $155 \times 10^6$  params. Both these figures share the same y-axis, and the grey dots highlight the centre of the blobs.



**Accuracy per parameter vs. network.** Information density (accuracy per parameters) is an efficiency metric that highlights the capacity of a specific architecture to better utilise its parametric space. Models like VGG and AlexNet are clearly oversized, and do not take full advantage of their potential learning ability. On the far right, ResNet-18, BN-NIN, GoogLeNet and ENet (marked by grey arrows) do a better job at “squeezing” all their neurons to learn the given task, and are the winners of this section.



## Oh no! It seems too DEEP!!!!



Source: <https://medium.com/@jacklindsai/can-a-neural-network-be-too-deep-7347fe461172>

# Next Time

**How to use Deep Neural Networks to do regression?**